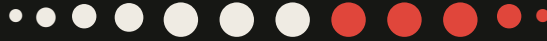


ENGLISH EDITION · 76 NODES



Cancerpitalism

Growth for growth's
sake

Cancer and capitalism,
mechanism by mechanism: a
book in 76 nodes.

Idea and co-
editing: Nu a
l'art and AI
2026

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Summary of the architecture

Script of 76 nodes — English version (29 September 2026)

How to read

The **C** stands for both **cancer** and **capitalism**.

Each node states a single idea that holds for both. When a shared word is not precise enough, the text says so explicitly: **the cell or the firm, the tumour or concentrated capital, the body or society**.

Each node has a centre and three readings:

- **Centre:** central message, question, short answer and the correspondence between the two vocabularies.
- **Right — Cancer:** the biological mechanism, explained with oncological precision.
- **Left — Capitalism:** the economic, social and political mechanism, with data and authors.
- **Below — Limits and sources:** where the comparison holds, where it fails, and where every claim comes from.

The central diagram adds no new theory: it makes visible, with a single graphic relation, what the central message states.

Writing rules

1. **One word, one meaning.** Glossary terms always mean the same thing.
2. **No figure without a source.** If a figure is a contested estimate, the text says so.
3. **The bite comes from the facts.** Criticism needs no adjectives: it is enough to set one mechanism beside the other.
4. **The person with cancer is never the metaphor.** The comparison is between two processes, not between a person and a system.
5. **The metaphor is unfair to cancer.** A cell does not decide; a firm, a bank or a government does. When the comparison fails, it almost always fails against capitalism.

Glossary of correspondences

- **Cell** ↔ firm, unit of capital or economic agent. Never the person.
- **Tissue** ↔ community, sector or territory.
- **Body or organism** ↔ society, economy or biosphere, depending on scale.
- **Proliferation** ↔ accumulation of capital.
- **Tumour** ↔ concentration of capital and power.
- **Tumour microenvironment** ↔ the infrastructure that sustains accumulation: state, law, credit, labour, care and nature.
- **Angiogenesis** ↔ building infrastructure that carries resources to the centre.
- **Mutation** ↔ innovation, deregulation or a change of rules that gives an edge for growth.
- **Clonal selection** ↔ competition that preserves the most expansive forms.
- **Immune system** ↔ countervailing powers: public regulation, unions, the press, independent science, social movements.
- **Immune evasion** ↔ opacity, tax avoidance, lobbying and regulatory capture.
- **Apoptosis** ↔ the orderly disappearance of firms, privileges or structures that no longer serve the whole.

- **Cachexia** ↔ impoverishment of the peripheries to feed the centre.
 - **Metastasis** ↔ expansion into other territories, markets, cultures, classes, languages and spheres of life.
 - **Homeostasis** ↔ social, economic and ecological balance.
 - **Remission** ↔ a period of containment in which symptoms fall but causes remain active.
 - **Relapse** ↔ the return of accumulation and inequality after a crisis or a reform.
 - **Local treatment** ↔ regulating a sector, a specific tax or breaking up a monopoly.
 - **Systemic treatment** ↔ transforming ownership, power and distribution.
-

Prologue — The question

Node 01 — Growth without brakes: what for?

CENTRAL MESSAGE

The C does not grow to get anywhere: it grows in order to keep growing.

QUESTION

Can growth last long when it has no purpose other than growing?

SHORT ANSWER

Cancer has already given its answer: left unchecked, it ends the body that contains it. Capitalism has not answered yet. This book asks the question.

CORRESPONDENCE

Autonomous neoplasia ↔ the M-C-M' circuit (money seeking more money)

● CANCER

A healthy cell divides for a reason: to grow an embryo, to close a wound, to replace worn-out cells. When the reason disappears, division stops. The classic definition of neoplasia, formulated by the pathologist Rupert Willis and still quoted in textbooks, says exactly this: a mass of tissue whose growth exceeds and is uncoordinated with

that of the normal tissues, and **persists in the same excessive manner after cessation of the stimuli which evoked the change.**

That is the mark of cancer: growth no longer answers any need of the body. It is not "bad" growth because it is fast, but because it has become independent of its function.

● CAPITALISM

Karl Marx distinguished two circuits. In the first, someone sells a commodity in order to buy another one they need: C–M–C. The circuit ends when the need is satisfied. In the second, someone invests money to obtain more: M–C–M'. This circuit never ends, because M' only makes sense if it is invested again to become M''. Marx wrote it without ambiguity: the circulation of money as capital is an end in itself, and therefore **has no limits.**

Aristotle had already separated *oikonomia*, the management of the goods needed to live, from *chrematistics*, the art of accumulating wealth, which in his view has no natural measure. Capitalism is the first system that turns chrematistics into the general rule of the economy.

LIMITS

The correspondence is structural: in both cases growth continues when it no longer serves any function. It does not imply that the causes are the same. In 1968 Edward

Abbey wrote that "growth for the sake of growth is a cancerous madness"; in 1969, that it is "the ideology of the cancer cell". This book tries to test rigorously whether the phrase was an exaggeration or a diagnosis.

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CENTRAL DIAGRAM

Diptych. On the left, piles of coins that double ($M \rightarrow M' \rightarrow M''$); on the right, cells that divide ($1 \rightarrow 2 \rightarrow 4 \rightarrow 8$). Neither series has an end point.

Node 02 — An old suspicion

CENTRAL MESSAGE

Comparing economic growth with cancer is not new. What is missing is doing it with precision.

QUESTION

Is the comparison an insult or a diagnosis?

SHORT ANSWER

It will only be a diagnosis if every similarity survives both an oncologist's and an economist's scrutiny.

CORRESPONDENCE

Hallmarks of cancer ↔ hallmarks of capitalism

● CANCER

Cancer is not one disease but more than two hundred. They all share three properties that set them apart from benign tumours: **uncontrolled growth**, **invasion** of neighbouring tissues and, often, **metastasis** to distant organs. A benign tumour can be large, but it limits itself and does not invade.

In 2000 Douglas Hanahan and Robert Weinberg summed up the biology of cancer in a list of "hallmarks": self-sufficiency in growth signals, insensitivity to brakes, resistance to cell death, limitless replication, the creation of blood vessels, invasion and metastasis. The 2011 and 2022 revisions added reprogrammed metabolism, immune evasion and inflammation. This list serves as the book's protocol: many nodes test whether one of these hallmarks also appears in the economy.

● CAPITALISM

Capitalism is an economic system in which the means of production are mostly privately owned, labour is bought and sold as a commodity, and production is organised to obtain a profit that must be reinvested. Its forms vary widely (liberal, social, state, financial, platform), but they share this core: **accumulation is the goal and competition is the engine.**

The comparison with cancer has a history. In 1955 the physician Alan Gregg wrote in *Science* that "the world has cancer and the cancer cell is man". In 1999 the philosopher John McMurtry published *The Cancer Stage of Capitalism*, describing the financialised economy as growth that reproduces itself without any life function.

LIMITS

Gregg, and later the anthropologist Warren Hern, pointed at the human species as the tumour. This book does not: the diagnosis is not aimed at humanity but at a historical way of organising the economy. Susan Sontag warned that illness metaphors can blame the sick and justify political violence. That is why the comparison must be precise or not be made at all.

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CENTRAL DIAGRAM

A timeline with four documents: Gregg (1955), Abbey (1968), McMurtry (1999) and this book (2026).

Node 03 — The C is born from within

CENTRAL MESSAGE

The C does not come from outside: it uses the normal circuits of the system itself and puts them to its own service.

QUESTION

How can a system feed what impoverishes it?

SHORT ANSWER

Because the **C** invents nothing new: it redirects mechanisms that already existed and that the body or society need.

CORRESPONDENCE

Oncogene (an altered normal gene) ↔ a normal institution redirected towards accumulation

● CANCER

Flu is a virus and tuberculosis a bacterium. Cancer is neither: it is **cells of the body itself** that have accumulated alterations in their DNA. The genes that drive cancer, oncogenes, are altered versions of normal

genes, proto-oncogenes, which control growth when a tissue needs repair. Cancer builds no new machinery: it takes over the machinery that was already there.

That is why it is so hard to treat. The immune system is prepared to recognise what is foreign. A tumour cell is, to a large extent, **one of the body's own cells**. Some viruses, such as human papillomavirus, can trigger cancer, but the process is still a transformation of the organism's own cells.

● CAPITALISM

Capitalism did not arrive from nowhere either. It was born inside societies that already had markets, money, credit, property law and trade. The historian Fernand Braudel described it as the top floor of a building: below lie material life and local markets; on top, a few actors who control the great flows and use markets to their own advantage.

Karl Polanyi showed that the nineteenth-century "free market" was not spontaneous: it was built by laws, the enclosure of common land and a state that turned land, labour and money into commodities. Capitalism uses institutions society needs (contracts, banks, courts, schools) and points them towards accumulation.

LIMITS

A mutation is a copying error or chemical damage. The redirection of institutions was a political process, with winners and losers. In both cases, however, the lesson is the same: **the enemy is not a foreign body**, and so the system does not recognise it as an enemy.

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CENTRAL DIAGRAM

Diptych. On the left, an ordinary building in a city grows and draws resources from its neighbours; on the right, a body with one of its own cells carrying altered DNA.

Node 04 — The patient is not the disease

CENTRAL MESSAGE

A person with cancer is not a metaphor for capitalism. Often they are its victim.

QUESTION

How can we compare two processes without turning the person who suffers into a symbol or a culprit?

SHORT ANSWER

By always separating the person from the disease, and by pointing at whoever manufactured the risk and whoever charges for the treatment.

CORRESPONDENCE

Blaming the patient's lifestyle ↔ blaming the consumer's individual choice

● CANCER

A significant share of cancers depends on chance: every cell division carries small copying errors, and tissues that divide more accumulate more risk. Another share depends on exposure to carcinogens: tobacco, alcohol, asbestos, air pollution, radiation, some viruses. The

World Health Organization estimates that between 30% and 50% of cancers could be prevented by reducing these risk factors.

None of these causes makes the sick person guilty. Speaking of "fighting", "winning" or "losing" against cancer suggests that whoever dies did not fight hard enough. Susan Sontag wrote her essay precisely against this moral burden.

- **CAPITALISM**

Many carcinogens are not natural accidents: they are products that someone manufactures, advertises and defends. The tobacco industry knew from the 1950s that its product caused cancer and funded decades of scientific doubt. A 1969 Brown & Williamson internal memo put it plainly: "Doubt is our product." The next strategy was to shift responsibility to the individual: "if you smoke, that's your choice".

Public health research calls this the **commercial determinants of health**: business products and practices that increase disease. In Europe, four industries (tobacco, alcohol, ultra-processed food and fossil fuels) are associated with some 2.7 million deaths a year. And when disease arrives, treatment also has a price: cancer is one of the leading causes of household financial ruin in countries without universal healthcare.

LIMITS

The patient is neither the tumour nor capital. The comparison is between processes, never between people. If anyone resembles the tumour, it is not the one who falls ill but the one who profits from the risk.

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CENTRAL DIAGRAM

A factory sends smoke and a pack of cigarettes towards a person with a spot on the lungs. The arrow goes from whoever manufactures the risk to whoever falls ill.

Node 05 — The cell does not decide; the board does

CENTRAL MESSAGE

If the comparison is unfair, it is unfair to cancer: a cell does not know what it is doing.

QUESTION

Where does the metaphor end?

SHORT ANSWER

Where minutes, laws, internal reports and signatures begin.

CORRESPONDENCE

Natural selection without intent ↔ a documented decision with foreseen consequences

● CANCER

A tumour cell does not want to grow, does not calculate and does not choose. Its genetic alterations are random. Those that confer an advantage for surviving and dividing multiply; the others disappear. Calling a cell "selfish", a "cheater" or an "invader" is a shortcut of language: it describes the result of selection, not a will.

That is why a tumour is responsible for nothing. It has not read any report, hidden any data or hired any law firm.

● CAPITALISM

Capitalism, by contrast, is made of decisions. In 1977 Exxon's scientists were already projecting global warming with a precision that a 2023 study published in *Science* has confirmed. The company chose to fund doubt. The tobacco industry chose the same. Limited liability, the legal personhood of corporations or patent protection are not natural facts: they are written rules, voted and defended in court.

The legal scholar Katharina Pistor puts it this way: **capital is coded in law**. Any asset becomes capital when the law grants it priority, durability and protection against others. What one law has coded, another can decode.

LIMITS

This is the fundamental limit of the metaphor, and it is an aggravating circumstance for capitalism. Cancer is a tragedy without an author; capitalism has authors, beneficiaries and archives.

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CENTRAL DIAGRAM

Diptych. On the left, a board of directors signs a sealed document; on the right, dice beside cells that mutate by chance.

Node 06 — When the disease is the environment

CENTRAL MESSAGE

One in five people will have cancer in their lifetime. The social C surrounds all of us, and we call it the economy.

QUESTION

How do you diagnose a disease that has become the norm?

SHORT ANSWER

By looking at it from outside what we have been taught to call normal.

CORRESPONDENCE

A disease affecting part of the body ↔ a system containing the whole of society

● CANCER

In 2024 about 20.6 million new cancers were diagnosed worldwide and there were about 9.8 million deaths. One in five people will develop one during their lifetime. That is a lot, but cancer is still an exception within each body: a part that has become deregulated, surrounded by tissues that still work.

Knowledge also advances. In the United States cancer mortality fell by 34% between 1991 and 2022, mainly thanks to less smoking, early detection and better treatments. Cancer can be named, measured and treated because we clearly distinguish it from health.

● CAPITALISM

With capitalism the opposite happens. Almost all of humanity lives inside markets, wages, debts and prices. It is not an exception: it is the atmosphere. Mark Fisher called it **capitalist realism**: the sense that there is no possible alternative, that "it is easier to imagine the end of the world than the end of capitalism". Margaret Thatcher said it in four letters: TINA, *There Is No Alternative*.

When a disease is the environment, its symptoms look natural: precarity is called flexibility; concentration, efficiency; ecological destruction, an externality.

LIMITS

Cancer is not a rare disease: the lifetime risk is one in five. But the difference in scale remains at the heart of the comparison: cancer is a deregulated part within a body; capitalism is the rule of the whole.

SOURCES

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CENTRAL DIAGRAM

Five people: one has cancer. Below, the same five people inside a price tag: everyone lives inside the market.

Part I

Cancer: Cellular deregulation

**Capitalism: Deregulated
accumulation**

Common idea: a part stops obeying the limits of the whole and turns its own reproduction into its only goal.

Chapter 1 — Cooperation, function and limit

Node 07 — Living is cooperating

CENTRAL MESSAGE

A body and a society exist because their parts cooperate. The C begins when one part stops cooperating and gains from it.

QUESTION

What turns millions of parts into a liveable whole?

SHORT ANSWER

Five agreements: do not grow more than needed, accept dying when necessary, divide the work, share the resources and look after the common environment.

CORRESPONDENCE

A cheating cell in a multicellular organism ↔ an agent that free-rides on the commons

● CANCER

For more than three billion years life was unicellular: each cell reproduced as much as it could. Multicellular organisms only became possible when cells accepted five restrictions: **limiting proliferation, accepting programmed death, dividing labour, sharing**

resources and maintaining the extracellular environment. The biologist Athena Aktipis describes cancer as the breakdown of these five agreements: a cell that "cheats" and goes back to behaving like a single-celled organism inside a body.

Cheating works in the short run. A cell that ignores the restrictions reproduces more than its neighbours. The problem is that its success depends on the body it is breaking.

- **CAPITALISM**

Societies also depend on cooperation: trust, shared norms, common goods, care and public infrastructure. Even the market needs most people not to cheat. Elinor Ostrom showed that many communities manage common goods for centuries with rules of cooperation.

Capitalism systematically rewards those who break these agreements without paying the cost: those who pollute the air everyone breathes, who avoid the taxes that pay for the roads they use, who pay wages people cannot live on and let the state make up the rest. In economics this is called **externalising costs**. In evolutionary biology it is called **cheating**.

LIMITS

A society is not an organism with a shared genome: its members have legitimately different interests. The comparison does not demand that everyone obey, only that no part be allowed to cheat against the whole with impunity.

SOURCES

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CENTRAL DIAGRAM

A tissue of neatly ordered hexagonal cells; in the middle, one cell breaks the pattern and proliferates.

Node 08 — Growth must serve a function

CENTRAL MESSAGE

Growth is healthy when it repairs, renews or meets a need. The C grows when there is nothing left to repair.

QUESTION

When does growth stop serving life?

SHORT ANSWER

When it continues after the need has been met.

CORRESPONDENCE

Reversible hyperplasia ↔ production for needs · Neoplasia ↔ production for profit

● CANCER

The body grows often and safely. When we cut ourselves, skin cells multiply until the wound is closed, and then they stop. During pregnancy the mammary glands grow; after breastfeeding they shrink again. This growth is called **hyperplasia**, and it has one essential property: **it is reversible**. When the stimulus disappears, the excess disappears.

Neoplasia is something else: growth no longer depends on the stimulus. It continues even though the wound is closed and the body does not need a single extra cell.

● CAPITALISM

Marx distinguished the **use value** of a thing (what it is for) from its **exchange value** (how much money it can fetch). An economy geared to needs stops when the need is covered. An economy geared to profit has no reason to stop: a satisfied need is a lost market.

In 1930 John Maynard Keynes calculated that, thanks to productivity, his grandchildren would work fifteen hours a week. Productivity grew more than he expected; the working week did not shrink. Growth stopped serving free time and began serving accumulation itself.

LIMITS

Not all economic growth beyond basic needs is useless: there is also knowledge, culture and health. The criterion is not quantity but whether growth answers a function or only itself.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a skin wound that closes and stops; on the right, cells that keep piling up with no wound at all.

Node 09 — Knowing when to stop

CENTRAL MESSAGE

A healthy part knows when to stop growing. For the C, stopping is a disease.

QUESTION

Why is a limit a form of health?

SHORT ANSWER

Because no function can occupy the space of all the others indefinitely.

CORRESPONDENCE

Contact inhibition and cell-cycle checkpoints ↔ the stationary state ·
Loss of brakes ↔ recession seen as illness

● CANCER

Healthy cells have several brakes. When they touch their neighbours they stop dividing: this is **contact inhibition**. Before each division, checkpoints verify that the DNA is intact; proteins such as p53 and Rb can halt the cycle. And every cell has a counter: after some 40 to 60

divisions the ends of the chromosomes (the telomeres) are too short and the cell stops dividing. This is the **Hayflick limit**.

Cancer cells lose these brakes one by one. One of their classic capabilities is replicative immortality: they reactivate telomerase and can divide indefinitely.

● CAPITALISM

In 1848 John Stuart Mill argued that a mature economy should reach a **stationary state**: no material growth, but moral, cultural and technical progress. A century later Joseph Schumpeter wrote that capitalism "is by nature a form or method of economic change and not only never is but never can be stationary".

Today a year of zero growth is not called stability: it is called a **recession**. Governments and central banks treat it as an emergency, because in a capitalist economy stopping means unemployment, bankruptcies and defaults. The system has no healthy resting state.

LIMITS

An economy that does not grow can also be unjust and unequal. The parallel does not claim that every halt is good, only that a system that cannot stop without falling into crisis has its brakes disabled.

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CENTRAL DIAGRAM

Diptych. On the left, a GDP gauge on which zero is marked as recession; on the right, chromosomes whose telomeres shorten, and a tumour chromosome that regenerates them with telomerase.

Node 10 — Knowing when to disappear

CENTRAL MESSAGE

In a healthy body, the parts that no longer serve die in an orderly way. The C is what refuses to die.

QUESTION

What happens when nothing agrees to give up space?

SHORT ANSWER

The part that does not die becomes a burden on the whole, and the whole pays to keep it alive.

CORRESPONDENCE

Evasion of apoptosis ↔ "too big to fail"

● CANCER

Every day between 50 and 70 billion cells die in an adult body. This is no accident: it is **apoptosis**, a programmed, clean and orderly death that removes old, damaged or surplus cells without inflaming the tissue. Without apoptosis the fingers of an embryo would not separate and genetic errors would accumulate.

Tumours learn to avoid it. They disable p53, the gene that orders damaged cells to die, or overproduce proteins such as BCL-2 that block it. The cell that should have died keeps dividing.

● CAPITALISM

Schumpeter celebrated "creative destruction": inefficient firms disappear and make room for new ones. But real capitalism practises selective apoptosis. It lets shops, workshops and jobs die, and keeps the biggest players alive. In 2008 banks that had caused the crisis were rescued with public money because they were "too big to fail". The Bank for International Settlements estimated that zombie firms (unable to pay the interest on their debt out of profits) rose from 2% of listed companies in the late 1980s to 12% in 2016.

Hereditary privileges work the same way: large fortunes that cross generations without having to serve any function.

LIMITS

Apoptosis is an internal programme of the cell; the closure of a firm has human victims. Arguing that certain structures must disappear does not mean abandoning the people who work in them, but ceasing to keep big capital alive with common resources.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, small shops closing and a big bank saved with a lifebuoy; on the right, cells breaking apart through apoptosis and a tumour cell that does not die.

Chapter 2 — The order to grow

Node 11 — When the brakes fail

CENTRAL MESSAGE

The C appears when stop signals lose their authority. In the case of capital, someone disconnected them.

QUESTION

What turns a normal activity into destructive expansion?

SHORT ANSWER

A stuck accelerator and disabled brakes.

CORRESPONDENCE

Oncogene (accelerator) and tumour suppressor gene (brake) ↔ incentives to grow and deregulation

● CANCER

To explain cancer, oncologists use the image of a car. **Oncogenes** are the accelerator: genes that, when altered, send constant growth signals. **Tumour suppressor genes** are the brakes: genes such as p53 or RB1 that halt division when something goes wrong. Losing a brake usually requires inactivating both copies of the gene; this is Alfred Knudson's "two-hit" hypothesis (1971).

A cancer usually needs both: a stuck accelerator and several broken brakes. A single alteration is not enough.

● CAPITALISM

The brakes on capital can also be disabled. After the 1929 crash, the United States separated commercial banking from investment banking with the Glass-Steagall Act (1933). It was repealed in 1999. The following year another law left financial derivatives unregulated. The 2008 crisis arrived eight years later. In Europe, the free movement of capital and tax competition between states played the same role: brakes dismantled one by one.

Here lies the decisive difference: the mutations of cancer are the product of chance. Deregulation was drafted, voted and celebrated.

LIMITS

Economic brakes do not disappear by mistake but by political decision. The correspondence describes the mechanism (accelerator plus broken brakes), not its origin.

SOURCES

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CENTRAL DIAGRAM

A car's pedals: the stuck accelerator (oncogene, incentives) and the brake with its cable cut (p53, Glass-Steagall).

Node 12 — Growing so as not to disappear

CENTRAL MESSAGE

The C grows because stopping means losing territory, nutrients or market.

QUESTION

What happens when standing still is the same as dying?

SHORT ANSWER

Expansion stops being an option and becomes an obligation.

CORRESPONDENCE

Clonal competition within the tissue ↔ the coercive laws of competition

● CANCER

A precancerous tissue contains many slightly different populations of cells. They all compete for the same resources: space, oxygen, glucose and growth factors. Those that divide more slowly are displaced by those that divide faster. No cell "wants" to win: those that do not proliferate simply leave fewer descendants and vanish from the population.

● CAPITALISM

Marx called this mechanism **the coercive laws of competition**: a capitalist who does not reinvest and expand production is overtaken by one who does, and ends up absorbed or ruined. It is not a matter of personal character; it is a structural condition.

Economists call this pressure the **growth imperative**. Mathias Binswanger showed that in a monetary economy with credit and interest, aggregate profits are only positive if the economy grows. Oliver Richters and Andreas Siemoneit have identified several mechanisms that reinforce it: competition, debt, shareholder expectations and the dependence of employment on growth.

LIMITS

A firm can decide not to grow and survive in a niche; a cell cannot decide anything. But the system penalises that decision, which is why exceptions do not change the rule.

SOURCES

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CENTRAL DIAGRAM

*A conveyor belt that speeds up, carrying a factory, a house and a tower;
the house that stops falls off the end.*

Node 13 — Competition selects expansion

CENTRAL MESSAGE

Not every part needs to want to grow: it is enough that those that do not grow disappear.

QUESTION

How does a system become expansive when nobody decided it?

SHORT ANSWER

Through selection: each round of competition removes the moderate forms and keeps the voracious ones.

CORRESPONDENCE

Clonal evolution ↔ market selection

● CANCER

In 1976 Peter Nowell proposed that a tumour evolves like a Darwinian population. Cells accumulate random mutations; the environment selects those that survive and divide most. Generation after generation, the tumour becomes enriched in more aggressive cells. This is **clonal evolution**: natural selection in miniature, working against the organism.

● CAPITALISM

In 1950 the economist Armen Alchian argued that entrepreneurs need not consciously maximise profit: the market selects the firms that, for whatever reason, make more of it, and eliminates the rest. Richard Nelson and Sidney Winter developed this idea into an evolutionary theory of the firm.

The result is the same as in a tumour. A cooperative that prioritises wages, a family business that does not want to grow or a prudent bank can be bought, displaced or ruined by more aggressive competitors. The system needs no villain: it needs a selection rule.

LIMITS

Market selection depends on written rules: taxation, competition law, labour standards. Change the rules and you change what is selected. In a tumour, the rules are the chemistry of the body.

SOURCES

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CENTRAL DIAGRAM

Three Petri dishes: many different colonies, then one that grows, and finally a single colony that takes over everything.

Node 14 — Accumulation gives the power to accumulate

CENTRAL MESSAGE

Every resource captured makes it easier to capture more.

QUESTION

Why does concentration accelerate instead of levelling off?

SHORT ANSWER

Because the C feeds on its own signals: more mass, more vessels; more capital, more power to obtain capital.

CORRESPONDENCE

Autocrine signalling ↔ a return on capital higher than growth ($r > g$)

● CANCER

A healthy cell needs other cells to send it growth factors before it divides. Many tumour cells make their own: they produce the signal and carry the receptor to receive it. This closed circuit is called **autocrine signalling**. What is more, the bigger a tumour, the more blood vessels it

induces and the more nutrients it receives. And genetic instability generates new variants faster and faster. Each advantage makes the next one easier.

● CAPITALISM

Thomas Piketty has documented that, in the long run, the return on capital (r) tends to exceed the growth rate of the economy (g). When $r > g$, fortunes grow faster than wages and wealth concentrates. The sociologist Robert Merton called this the **Matthew effect**: "unto every one that hath shall be given".

The figures confirm it: according to the World Inequality Report 2026, the richest 10% own three quarters of the world's wealth, and the poorest half, 2%. Some 56,000 people (the top 0.001%) hold three times more wealth than half of humanity; their share has risen from 3.7% in 1995 to 6.1% in 2025. A 2011 analysis of 43,000 multinationals found that 147 companies controlled 40% of the value of the whole network.

LIMITS

Economic concentration has possible counterweights (taxes, inheritance rules, competition law) that the body lacks against a tumour. That $r > g$ is a tendency does not make it an inevitable law: between 1914 and 1970, wars, crises and fiscal policies reversed it.

SOURCES

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CENTRAL DIAGRAM

A snowball rolling downhill and growing bigger and bigger, with coins and cells inside it.

Node 15 — A finite world under an infinite command

CENTRAL MESSAGE

The C demands indefinite growth inside a finite body and a finite planet.

QUESTION

What happens when the command to grow recognises no boundary?

SHORT ANSWER

The boundary returns anyway: as necrosis, exhaustion, pollution or collapse.

CORRESPONDENCE

Tumour growth limited by the host ↔ economic growth limited by the biosphere

● CANCER

No tumour can grow indefinitely. The classic mathematical models (the Gompertz curve) show that growth slows as the tumour gets bigger: it runs short of oxygen, space and nutrients, and the centre of the mass dies of necrosis. But the tumour does not stop out of prudence: it stops because the body is exhausted. When the host dies, the tumour dies too.

There is an unsettling exception: a few cancers have become transmissible. Canine transmissible venereal tumour has been passing from one dog to another for some 11,000 years; Tasmanian devil facial tumour disease and some cancers of bivalve molluscs do the same. They are the only known cases in which the tumour outlives its host: it finds new bodies in which to carry on.

● **CAPITALISM**

An economy growing at 3% a year doubles in size every 23 years. David Harvey points out that this is the rate capital considers "healthy". In 1972 the report *The Limits to Growth* showed that exponential growth in production and consumption would collide with the planet's physical limits during the twenty-first century.

In 2025 the Planetary Health Check concluded that seven of the nine planetary boundaries have already been crossed: climate, biosphere integrity, land use, freshwater, the nitrogen and phosphorus cycles, novel chemical entities and ocean acidification. The plans to colonise Mars, promoted by some of the richest men in the world, are the economic version of transmissible cancer: find a new body once the first one is used up.

LIMITS

The economy can grow in value without growing in matter (services, knowledge), but the data show that absolute decoupling of GDP from resource use is rare, local and far too slow to meet climate targets.

SOURCES

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CENTRAL DIAGRAM

A planet with a growth curve bursting out of it, and a rocket heading for another planet.

Chapter 3 — What is the capital that makes us ill?

Node 16 — Not all exchange is C

CENTRAL MESSAGE

Buying, selling and exchanging are not, in themselves, capitalism; growing is not, in itself, cancer.

QUESTION

When does exchange become an invasive logic?

SHORT ANSWER

When all circulation is subordinated to accumulating more capital, and when that accumulation breaks through the limits that contained it.

CORRESPONDENCE

Benign tumour ↔ local market · Malignant tumour ↔ capitalist accumulation

● CANCER

Not every mass is cancer. A **benign tumour** (a lipoma, a fibroma, a small polyp) grows, but stays encapsulated, does not invade neighbouring tissues and does not metastasise. It may be a nuisance or press on something, but it rarely kills. What defines **malignancy** is not size but the capacity to break limits: to invade and to spread.

● CAPITALISM

Markets have existed for thousands of years. For most of history, however, they were embedded in societies that limited their reach: religious rules against usury, guilds, common land, just prices. Karl Polanyi called what happened in the nineteenth century **disembedding**: the market stopped being a part of society, and society began to function as an appendage of the market.

Marx put it with the two circuits: C–M–C (selling in order to buy what you need) is exchange; M–C–M' (investing to obtain more money) is capital. The first formula closes; the second starts again indefinitely.

LIMITS

The border between market and capitalism is not as sharp as the one between benign and malignant tumours. But the distinction allows capitalism to be criticised without idealising the absence of exchange.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, selling an apple to buy bread (the circuit closes); on the right, money passing through a factory and returning as more money,

in a loop.

Node 17 — Capital is a relation

CENTRAL MESSAGE

Capital is not just money, just as a tumour is not just a mass of cancer cells.

QUESTION

What gives an accumulation real power?

SHORT ANSWER

Control over others: their resources, their infrastructure and the conditions in which they live.

CORRESPONDENCE

The tumour as an organ (tumour cells + recruited stroma) ↔ capital as a social relation

● CANCER

For decades the tumour was studied as a bag of malignant cells. Today we know that a tumour is an **organ** made up of many cell types: cancer cells, but also fibroblasts, blood vessels, immune cells and an extracellular matrix that holds them together. Douglas Hanahan and Lisa Coussens called these recruited cells

"accessories to the crime". Much of a tumour's power comes from the relations it imposes on the normal cells around it.

● CAPITALISM

Marx explained it with a colonial example. A certain Mr Peel took means of subsistence and production worth £50,000, and 3,000 workers, to Australia. On arrival the workers left to farm free land, and Mr Peel was left without a servant to make his bed or fetch him water. Marx's conclusion: **capital is not a thing, but a social relation between persons, established by the instrumentality of things**. Money is only capital when someone is obliged to work for whoever owns it.

That is why capital defends so fiercely the ownership of land, housing, patents and data: it does not control objects, it controls other people's access to what they need.

LIMITS

Stromal cells cannot refuse to cooperate. Workers, tenants and users can organise, and the history of capitalism is also the history of that resistance.

SOURCES

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CENTRAL DIAGRAM

An Australian beach in the nineteenth century: a ship, piles of money and goods, and the workers walking off towards free land.

Node 18 — The system can do harm without a single mind

CENTRAL MESSAGE

The C needs no central command to order each harm.

QUESTION

How can a structure coordinate destructive behaviour without anyone coordinating it?

SHORT ANSWER

By rewarding expansion and punishing whoever stops or cooperates.

CORRESPONDENCE

Emergent tumour behaviour ↔ the invisible hand

● CANCER

A tumour has no brain and no plan. No cell gives orders to the others. Even so, the whole behaves as if it had a strategy: it grows vessels, deceives the immune system, invades and colonises. This "intelligence" is **emergent behaviour**: the sum of millions of cells responding locally to their surroundings, filtered by selection.

● CAPITALISM

Adam Smith wrote that each individual, pursuing his own interest, is "led by an invisible hand to promote an end which was no part of his intention". Smith was thinking of the common good. The same invisible hand can lead to common harm.

In 1970 Milton Friedman said it without metaphors: "the social responsibility of business is to increase its profits". If every board fulfils this mandate, none of them has decided to heat the planet, drain the aquifers or widen inequality. Together, however, they do it with perfect coordination.

LIMITS

Here the metaphor lets capitalism off too lightly: within the system there are very powerful actors who do coordinate (lobbies, business associations, summits). The harm is emergent and, at the same time, organised.

SOURCES

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CENTRAL DIAGRAM

A shoal of fish which, without any fish leading it, forms a single direction.

Node 19 — Debt compels growth

CENTRAL MESSAGE

Debt turns the future into an obligation: to produce more tomorrow to pay for what was spent today.

QUESTION

Why can an indebted economy not stop?

SHORT ANSWER

Because its balance depends on growth. If growth stops, the whole system collapses.

CORRESPONDENCE

Oncogene addiction ↔ dependence on growth to service debt

● CANCER

Many tumour cells depend on a single oncogene to survive. This is **oncogene addiction**, described by Bernard Weinstein in 2002. The cell has reorganised its entire functioning around that growth signal; if the signal suddenly stops, the cell does not return to normal: it dies.

Some of the most effective drugs, such as imatinib for chronic myeloid leukaemia, work precisely by cutting that signal.

● CAPITALISM

A credit-based economy lives with a similar addiction. Total world debt (public and private) stood at 251 trillion dollars in 2024, more than 235% of world GDP, according to the International Monetary Fund. Repaying that debt with interest requires future incomes higher than present ones. If the economy stops growing, debts cannot be paid, banks stop lending, firms close and prices fall, which makes debts heavier still. Irving Fisher called this spiral **debt deflation** (1933). Hyman Minsky added a paradox: stability breeds instability, because calm periods encourage ever more borrowing.

LIMITS

Debt is not pathological in itself: it finances hospitals, schools and energy transitions. The problem is a system in which only growth makes it possible to repay.

SOURCES

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CENTRAL DIAGRAM

A drip bag marked "+3%" feeds a city drop by drop; if the tube is cut, everything falls.

Node 20 — Rent makes property grow

CENTRAL MESSAGE

The C can grow without producing anything: it is enough to stand where other people's resources pass.

QUESTION

How can property feed itself without making anything new?

SHORT ANSWER

By charging a toll: rents, interest, patents, fees and data.

CORRESPONDENCE

The tumour as a "nitrogen trap" ↔ rentier capitalism

● CANCER

In 1951 the physician George Mider observed that tumours capture nitrogen (the basic element of proteins) even when the patient is malnourished. The tumour takes it from the diet and, if that is not enough, from the proteins in the host's muscles. Mider called the tumour a **nitrogen trap**: an organ that absorbs resources with priority, regardless of the body's needs.

● CAPITALISM

A rent is income obtained by controlling a scarce asset, not by producing anything new. The owner of a flat is paid because others need to live in it; the holder of a patent, because others need the medicine; a digital platform, because shops and users cannot do without it. The economist Brett Christophers has described contemporary Britain as **rentier capitalism**, in which assets, not production, decide who gets rich.

Adam Smith already said that landlords "love to reap where they never sowed". Rent takes priority over needs: the rent is paid before the food.

LIMITS

Some income from assets is legitimate (small savings, a pension). The problem is the concentration of assets and the power to set the toll.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a toll booth on a road; on the right, a tumour capturing nitrogen atoms from all over the body.

Interlude — My political anatomy

A life is not a file, a cost or a productive unit.

Diagnosis and economics describe part of a life: a stage, a prognosis, a pay grade. Neither can replace biography, bonds, fear, desire or one's own voice.

This book compares processes; people are left out of the comparison.

Part II

Cancer: Tumour microenvironment

Capitalism: Infrastructure of accumulation

Common idea: the C does not grow alone; it needs flows, energy, labour, protection and an environment that sustains it or fails to stop it.

Chapter 4 — Who feeds the C?

Node 21 — The C is not alone

CENTRAL MESSAGE

The C is not an isolated mass: it is a network that depends on everything around it.

QUESTION

Can it grow on its own?

SHORT ANSWER

No. It needs a soil that feeds it, protects it or fails to stop it.

CORRESPONDENCE

Seed and soil ↔ capital and its background conditions

● CANCER

In 1889 the surgeon Stephen Paget wondered why breast cancer spread to some organs and not to others. His answer was an agricultural image: the tumour cell is a **seed** that only grows in suitable **soil**. A century later, biology has proved him right. The **tumour microenvironment** (vessels, fibroblasts, immune cells, matrix, chemical signals) largely decides whether an altered cell stays dormant or forms a tumour.

● CAPITALISM

Capital does not grow in a vacuum either. It needs a state that guarantees property and contracts, roads and ports, schools that train workers, families that feed and care for them, and a nature that supplies raw materials and absorbs waste. The philosopher Nancy Fraser calls all this the **background conditions** of capitalism: what the system needs, does not pay for and tends to exhaust.

That is why she calls it **cannibal capitalism**: a system that devours the base on which it depends.

LIMITS

The tumour microenvironment has no interests of its own: it is co-opted. The social background conditions are people and institutions that can stop cooperating. In a society, the soil can go on strike.

SOURCES

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CENTRAL DIAGRAM

A seed in a cross-section of soil; its roots reach credit, law, labour, care and nature.

Node 22 — Capturing the flows

CENTRAL MESSAGE

The C diverts to itself the flows that used to feed the whole.

QUESTION

How does it turn shared resources into its own fuel?

SHORT ANSWER

By placing itself at the crossing point of the flows and consuming them before they reach anyone else.

CORRESPONDENCE

Glucose capture by the tumour ↔ capture of financial flows

● CANCER

Many tumour cells consume glucose at a far higher rate than normal cells. This voracity is so characteristic that medicine uses it for diagnosis: **positron emission tomography (PET)** injects radioactively labelled glucose, and the tumour lights up because it hoards more of it than the rest of the body. The tumour gives itself away through its hunger.

● CAPITALISM

In today's economies a growing share of profit comes from controlling flows, not from producing. Three asset managers (BlackRock, Vanguard and State Street) are together the largest shareholder in about 88% of the companies in the S&P 500 index. The big digital platforms sit between seller and buyer and take a commission. Fund managers, big banks and platforms are the crossing point.

If there were an economic PET scan, it would show financial centres and tax havens lit up: captures of flow out of all proportion to their population and real output.

LIMITS

Capturing glucose is a biochemical property; controlling financial flows is a power regulated by laws that could be different.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a globe with bright spots over financial centres; on

the right, a body in a PET scan with the tumour lit up.

Node 23 — Building supply lines

CENTRAL MESSAGE

The tumour grows vessels; capital builds transport, credit and energy networks. Both do it to carry resources to the centre.

QUESTION

How does the C secure its future supply?

SHORT ANSWER

By building or capturing the channels that bring resources to it.

CORRESPONDENCE

Tumour angiogenesis ↔ extraction infrastructure

● CANCER

A tumour cannot grow beyond one or two millimetres without new blood vessels: oxygen cannot diffuse any further. In 1971 Judah Folkman proposed that tumours produce signals (such as the VEGF factor) that force the body to build vessels for them. This is **tumour angiogenesis**. These vessels, however, are chaotic:

tortuous, leaky and badly connected. They feed the tumour and, at the same time, open escape routes to the rest of the body.

● CAPITALISM

Colonial infrastructure has the same shape. Walter Rodney observed that African railways did not connect cities with each other: they ran from mines and plantations to ports. Eduardo Galeano gave his book on Latin America a vascular title: *Open Veins of Latin America*. Today, pipelines, logistics corridors and submarine cables often follow the same route: from the periphery that extracts to the centre that consumes.

Like tumour vessels, these infrastructures are not designed to irrigate the territory they cross, but to drain it.

LIMITS

Infrastructure can change its use and serve the territory it crosses; many railways and power grids do. The problem is not building channels, but whom they serve.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a territory with railway lines running from the mines to the port; on the right, a tumour with blood vessels converging on it.

Node 24 — The state can feed or limit

CENTRAL MESSAGE

The state can brake the C, but it can also feed it.

QUESTION

When does it act as a defence and when as a supply line?

SHORT ANSWER

It depends on who controls laws, taxes, bailouts, subsidies and public procurement.

CORRESPONDENCE

Hormone-dependent tumours ↔ capital dependent on subsidies and bailouts

● CANCER

Some cancers depend on signals produced by the body itself. Roughly seven out of ten breast cancers carry oestrogen receptors, and most prostate cancers depend on androgens. The hormonal system, which regulates normal functions, feeds the tumour. In 1941 Charles Huggins showed that lowering male hormones made prostate cancer regress; in 1966 he received the Nobel

Prize. Today **hormone therapy** is one of the most effective treatments: it does not attack the tumour, it cuts the signal the body was sending it.

● CAPITALISM

The state also sends growth signals to capital. According to the International Monetary Fund, fossil-fuel subsidies reached 7 trillion dollars in 2022, 7.1% of world GDP, once the environmental and health costs that producers do not pay are included. In 2008 states rescued banks with public money. Public procurement, tax exemptions and patents protected by law are growth hormones too.

But the state can also be the brake: competition law, progressive taxes, environmental regulation and public services limit accumulation.

LIMITS

Hormones cannot be voted on; subsidies can. Economic "hormone therapy" (withdrawing subsidies from what does harm) is possible and needs no scientific discovery, only a decision.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, an institutional building with a thick pipe of subsidies to an oil derrick and a small brake; on the right, a gland sending hormones to a tumour:

Node 25 — Capturing the defences

CENTRAL MESSAGE

The C protects itself by turning its watchers into allies.

QUESTION

How does it avoid being recognised and stopped?

SHORT ANSWER

By reprogramming the defences: the immune cells that should attack it look after it; the regulators who should control it protect it.

CORRESPONDENCE

Tumour-associated macrophages and regulatory T cells ↔ regulatory capture and revolving doors

● CANCER

The immune system can recognise and eliminate cancer cells. But many tumours do not just hide: **they recruit the defences**. Macrophages, cells that ought to destroy the tumour, are reprogrammed to produce growth factors and vessels. Regulatory T cells, which normally prevent

attacks on the body itself, gather in the tumour and damp down the immune response there. The watchers work for what they should be watching.

● CAPITALISM

In 1971 the economist George Stigler (Nobel Prize 1982) formulated the theory of **regulatory capture**: the agencies that should regulate an industry end up serving it, because the industry has more information, more resources and more interest than anyone else in what the regulator decides. **Revolving doors** (regulators who go on to work for the companies they regulated, and vice versa) are its most visible mechanism.

The 2008 crisis revealed regulators who had trusted the banks to regulate themselves, and rating agencies paid by the very issuers they rated.

LIMITS

Macrophages draw no salary; regulators do. Economic capture is a form of legalised corruption that could be banned with conflict-of-interest rules, transparency and independent funding.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a revolving door between the regulator and the company; on the right, a macrophage feeding the tumour instead of attacking it.

Node 26 — Crisis feeds it too

CENTRAL MESSAGE

A wound that does not close or a crisis that does not end is fertile ground for the C.

QUESTION

Why can instability concentrate more power?

SHORT ANSWER

Because inflammation and urgency weaken controls, cheapen resources and open up new spaces.

CORRESPONDENCE

Chronic inflammation ↔ the shock doctrine

● CANCER

In 1863 Rudolf Virchow already observed inflammatory cells inside tumours. In 1986 Harold Dvorak defined tumours as "**wounds that do not heal**": the tumour permanently activates the healing programme (proliferation, new vessels, tissue remodelling) and keeps it switched on. **Chronic inflammation** raises the risk of cancer: chronic hepatitis, ulcerative colitis and *Helicobacter pylori* infection are well-known examples.

● CAPITALISM

Milton Friedman wrote it in 1982: "Only a crisis—actual or perceived—produces real change." Naomi Klein called **the shock doctrine** the strategy of exploiting disasters (coups, wars, hurricanes, financial crashes) to push through privatisations and cuts that would be rejected in normal times.

The pandemic figures illustrate it: according to Oxfam, between March 2020 and November 2021 the ten richest men in the world doubled their fortunes, while 99% of humanity lost income. Crisis is an open wound, and the C is what grows best in it.

LIMITS

Inflammation is a physiological process; crises have causes and people responsible. But both reinforce the same pattern: a repair system that is permanently switched on ends up feeding what it should be repairing.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, the fortune of the ten richest men doubles between 2020 and 2021; on the right, an inflamed wound that does not close.

Chapter 5 — Metabolism, labour and energy

Node 27 — The C changes metabolism

CENTRAL MESSAGE

The C does not just grow more: it consumes differently — fast, wastefully and toxically for its surroundings.

QUESTION

Why is it not enough to look at how much it grows?

SHORT ANSWER

Because we also have to look at what it consumes, how it consumes it and what it leaves behind.

CORRESPONDENCE

The Warburg effect ↔ metabolic rift and the economy of waste

● CANCER

A normal cell burns glucose with oxygen and obtains between 30 and 36 molecules of energy (ATP). In 1924 Otto Warburg observed that many tumour cells ferment it even when oxygen is available: they get only 2 molecules of ATP per glucose. This is the **Warburg effect**. It seems absurd, but it makes sense for something that

only wants to grow: it is a fast metabolism that frees carbon atoms to build new cells. The waste product, lactic acid, is dumped into the surroundings, acidifies them and damages neighbouring tissue, which makes invasion easier.

● CAPITALISM

Marx spoke of a **metabolic rift** between society and the land: the industrial city took nutrients from the countryside as food and dumped them into the river as waste, never returning them to the soil. The modern economy keeps this logic. In 2022 the world wasted 1.05 billion tonnes of food, about 19% of what was available to consumers, while hundreds of millions of people went hungry.

Like the tumour, capitalism prefers a fast, inefficient metabolism if it generates more turnover: produce, sell, throw away and produce again.

LIMITS

The Warburg effect has real biological advantages for proliferation; economic waste often has none, apart from private profit. On this point, cancer is more rational than the market.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a bin full of discarded food; on the right, two bars: 36 ATP for the healthy cell and 2 ATP for the tumour.

Node 28 — Labour is social energy

CENTRAL MESSAGE

The C puts the healthy parts around it to work for it.

QUESTION

Who really produces the value that is accumulated?

SHORT ANSWER

The people who work, those who care for them and the ecosystems that supply the matter. Capital organises the appropriation.

CORRESPONDENCE

Cancer-associated fibroblasts ("reverse Warburg effect") ↔ surplus value and wage labour

● CANCER

Fibroblasts are cells that maintain the structure of tissues. Inside a tumour they are reprogrammed into cancer-associated fibroblasts, which build matrix, growth factors and vessels for the tumour. In 2009 a research group proposed the **reverse Warburg effect**: in some tumours it is the fibroblasts that ferment glucose and

hand lactate and other nutrients to the cancer cells, which burn them to grow. The healthy cells do the work and the tumour collects the energy.

● CAPITALISM

Marx described capital as **dead labour that, vampire-like, only lives by sucking living labour**, and lives the more, the more labour it sucks. The mechanism is **surplus value**: in a working day the worker produces more value than they are paid in wages, and the difference belongs to the owner.

The recent trend is clear: since the 1980s the share of national income going to wages has fallen in most countries, while the share going to capital has grown. Productivity has risen; what the worker keeps of it, much less so.

LIMITS

The labour theory of value is disputed among economists, but the fall of the wage share is a measured fact. Unlike fibroblasts, workers can go on strike.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, an eight-hour working day split between wages and surplus value; on the right, a fibroblast passing energy to a tumour cell.

Node 29 — Fossil energy accelerates the C

CENTRAL MESSAGE

An excess of cheap energy does not make health grow: it makes grow whatever was already primed to grow.

QUESTION

What made it possible to expand production, transport and extraction on this scale?

SHORT ANSWER

Burning in two centuries an energy the Earth had stored for hundreds of millions of years.

CORRESPONDENCE

Excess energy (obesity, insulin) as a risk factor ↔ fossil fuels as an accelerator of accumulation

• CANCER

Chronic excess energy in the body is a recognised cancer risk factor. In 2016 a working group of the International Agency for Research on Cancer (IARC) concluded that excess body fat increases the risk of at least thirteen types of cancer. One of the mechanisms is hormonal: high levels of insulin and of the IGF-1 factor stimulate cell

proliferation. Abundant energy does not create cancer, but it feeds the cells already primed to grow without brakes.

● CAPITALISM

Andreas Malm has shown that nineteenth-century British industry did not switch to the steam engine because coal was cheaper than water, but because it allowed workers to be concentrated in towns and controlled more easily. Fossil energy accelerated accumulation: more production, more transport, more extraction.

Those who burned this energy have names. A 2014 study attributed 63% of industrial emissions of CO₂ and methane between 1751 and 2010 to just 90 producers of fossil fuels and cement. Between 2016 and 2022, after the Paris Agreement, 57 producers generated 80% of global fossil emissions.

LIMITS

A good share of these producers are state-owned companies or states (Saudi Arabia, Russia, China). Fossil dependence is not exclusive to private capitalism; it belongs to any economy organised around growth.

SOURCES

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CENTRAL DIAGRAM

An oil derrick drawing from an ancient fossil deposit; an emissions curve rises steeply out of it.

Node 30 — Expelling the damage

CENTRAL MESSAGE

The C protects its centre by dumping waste, pain and precarity on the peripheries.

QUESTION

Who bears the costs that never appear on the balance sheet?

SHORT ANSWER

The weakest tissues, the least protected workers, the poorest territories and the generations not yet born.

CORRESPONDENCE

Acidification of the microenvironment ↔ cost externalisation

● CANCER

Tumour cells pump protons and lactic acid outwards to keep their interior in optimal condition. The result is an acidic environment (a pH of 6.5 to 6.9, when normal is 7.4) that damages neighbouring healthy cells but that tumour cells tolerate. Robert Gatenby proposed that this

acidification is a weapon of invasion: the tumour destroys the neighbourhood with its waste and occupies the space left behind.

● CAPITALISM

Economists call **externalities** the costs an activity imposes on third parties without paying for them. Karl William Kapp was more direct in 1950: capitalism is a system of **cost-shifting**. In 1991 an internal World Bank memo signed by Lawrence Summers put it in writing: "I think the economic logic behind dumping a load of toxic waste in the lowest wage country is impeccable."

The data confirm it: the richest 10% of the world's population are responsible for half of emissions, yet heatwaves, droughts and floods hit mostly the poorest half, responsible for only 8%.

LIMITS

Summers later said the memo was ironic. The logic it describes, however, works every day with no need for irony.

SOURCES

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CENTRAL DIAGRAM

A ship loaded with toxic barrels leaves a clean city for a poor coast.

Node 31 — The invisible holds up the system

CENTRAL MESSAGE

The C depends on care and regeneration work that it neither pays for nor counts.

QUESTION

What is left out when we only measure what is sold?

SHORT ANSWER

Child-rearing, care, housework, community health and the regeneration of ecosystems.

CORRESPONDENCE

The host's homeostasis that sustains the tumour ↔ unpaid reproductive labour

● CANCER

A tumour has no liver, kidneys or lungs. It lives because the rest of the body keeps doing its job: the liver removes toxins, the kidneys regulate the blood, the heart pumps it and the lungs oxygenate it. This **homeostasis** is the invisible work that keeps the tumour's living conditions in place. The tumour contributes nothing to it; on the contrary, it wears it down.

● CAPITALISM

According to the International Labour Organization, 16.4 billion hours of unpaid care work are performed every day worldwide, three quarters of them by women. Oxfam has valued it at no less than 10.8 trillion dollars a year, three times the value of the entire global tech industry. None of these hours appears in GDP.

Silvia Federici and Maria Mies showed that this work was **feminised and made invisible** precisely when capitalism was born. Nancy Fraser sums it up: capital needs people who have been raised, fed and cared for, but does not want to pay for their production.

LIMITS

The body cannot stop performing homeostasis; the people who care can go on strike, as they have done in the feminist strikes of 8 March.

SOURCES

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CENTRAL DIAGRAM

An iceberg: above the water, what GDP counts; below it, cooking, raising children and caring.

Node 32 — Centres fed by peripheries

CENTRAL MESSAGE

The C makes some centres grow by consuming the peripheries.

QUESTION

What geography does accumulation produce?

SHORT ANSWER

Hypertrophied centres and peripheries forced to supply matter, energy, labour and their own flesh.

CORRESPONDENCE

Tumour cachexia ↔ unequal exchange and the drain from the Global South

● CANCER

Cachexia is a wasting syndrome: the patient loses muscle and fat even while eating. The tumour and the inflammation it provokes send signals (cytokines such as IL-6) that order the body to break down its own peripheral tissues. Muscle proteins become fuel for the tumour and for the inflammatory response. Cachexia affects up to eight in ten patients with advanced cancer

and is estimated to be directly responsible for a significant share of deaths: the body consumes itself to feed the centre.

● CAPITALISM

World-systems theorists (Samir Amin, Arghiri Emmanuel, Immanuel Wallerstein) described a world economy in which the prices of the South are systematically lower than those of the North for the same amount of labour and nature. A 2022 study measured the volume of this **unequal exchange**: in 2015 rich countries made a net appropriation from the Global South of 12 billion tonnes of materials, 822 million hectares of land, 21 exajoules of energy and 188 million person-years of labour, worth more than 10 trillion dollars at Northern prices. Every year.

LIMITS

The peripheries are not passive tissues: they have states, movements and elites of their own that often take part in the drain. But the net direction of the flow is measurable and persistent.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, arrows running from South to North; on the right, the muscle of an arm emptying to feed the tumour.

Chapter 6 — Manufacturing demand and occupying attention

Node 33 — Producing needs

CENTRAL MESSAGE

The C does not wait for the whole to need it: it manufactures the demand that feeds it.

QUESTION

Do we desire because we lack something, or do we lack something because someone needs us to desire?

SHORT ANSWER

The system turns human impulses into repeatable, saleable shortages.

CORRESPONDENCE

Paracrine signalling that recruits the host ↔ the dependence effect and advertising

● CANCER

A tumour does not wait for the body to send it resources: it asks for them. It secretes chemical signals (such as PDGF, TGF- β or VEGF) that recruit fibroblasts, immune cells and vessels, and order them to produce more growth factors. This is **paracrine signalling**: the tumour manufactures the body's demand for itself. Healthy tissues end up working to satisfy a need that is not theirs.

● CAPITALISM

In 1958 John Kenneth Galbraith called the **dependence effect** the fact that production itself creates the wants it then satisfies: "If production creates the wants it seeks to satisfy, [...] then the urgency of the wants can no longer be used to defend the urgency of the production." Edward Bernays, Sigmund Freud's nephew and the father of public relations, had already used psychoanalysis to sell cigarettes to women as "torches of freedom" (1929).

In 2024 global advertising spending passed one trillion dollars for the first time. An entire industry devoted to manufacturing needs.

LIMITS

Not all desires are manufactured, and advertising also informs. The parallel focuses on the mechanism: what grows actively produces the demand for its own growth.

SOURCES

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CENTRAL DIAGRAM

A billboard saying "You need this" in front of a person.

Node 34 — Buying an identity

CENTRAL MESSAGE

The C disguises itself as a useful cell; the commodity disguises itself as an identity.

QUESTION

What are we buying when we buy more than an object?

SHORT ANSWER

A provisional image of ourselves, which expires with the season.

CORRESPONDENCE

Phenotypic plasticity and dedifferentiation ↔ conspicuous consumption and brand identity

● CANCER

Healthy cells have a stable identity: a liver cell is a liver cell. Many tumour cells lose this identity (**dedifferentiation**) and can take on others: they recover embryonic programmes, change shape to migrate or even imitate blood vessels (so-called vasculogenic mimicry). In 2022 Hanahan added **phenotypic plasticity** to the list of cancer hallmarks: changing identity is a way of surviving.

● CAPITALISM

In 1899 Thorstein Veblen described **conspicuous consumption**: buying things to display status, not to use them. Jean Baudrillard added that in consumer society we do not buy objects but signs. The brand does not sell a trainer: it sells belonging, youth, rebellion or success.

Identity becomes a commodity with a sell-by date. And capitalism also changes its identity when it suits it: it presents itself as green, feminist, local or caring, depending on what the market wants.

LIMITS

This correspondence is weaker than the others. Tumour plasticity is a survival mechanism with no symbolic dimension. What they share is an unstable identity in the service of expansion.

SOURCES

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CENTRAL DIAGRAM

A trainer with tags hanging from it: status, youth, rebellion, success.

Node 35 — Making what still works expire

CENTRAL MESSAGE

The C removes what still works in order to take its place.

QUESTION

Why must what is useful be made to look inadequate?

SHORT ANSWER

Because a thing that lasts is a sale that is not repeated.

CORRESPONDENCE

Cell competition and "super-competitors" ↔ planned obsolescence

● CANCER

Tissues display a phenomenon called **cell competition**: cells compare their fitness with that of their neighbours and the least fit are eliminated. It is a useful quality control. In 2004 researchers discovered that cells with high levels of the MYC oncogene become "**super-competitors**": they induce the death of perfectly healthy neighbouring cells and take their space. The eliminated cells were working fine; they were simply less voracious.

● CAPITALISM

In 1924 the big light-bulb manufacturers (Osram, Philips, General Electric and others) formed the **Phoebus cartel** and agreed to limit bulb life to 1,000 hours; anyone making longer-lasting bulbs was fined. In 1932 Bernard London proposed **planned obsolescence** as a policy to end the Great Depression. Vance Packard criticised it in 1960.

Today the European Union has had to pass a right-to-repair directive (2024) because manufacturers of phones, household appliances and computers were systematically making repair difficult.

LIMITS

Cell competition is also a health mechanism that removes damaged cells. The problem, in both cases, is when the selection criterion stops being function and becomes the capacity to displace.

SOURCES

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CENTRAL DIAGRAM

A light bulb, a 1,000-hour clock and a bin: the bulb goes into it still working.

Node 36 — Colonising attention

CENTRAL MESSAGE

The C turns time, gaze, sleep and conversation into extractable resources.

QUESTION

Which part of us remains outside the market?

SHORT ANSWER

Less and less: even night and rest produce data and revenue.

CORRESPONDENCE

Circadian disruption (probably carcinogenic) ↔ the 24/7 economy and surveillance capitalism

● CANCER

The body has internal clocks that regulate cell division, DNA repair and the secretion of hormones such as melatonin. Disrupting them has consequences. Since 2007 the IARC has classified **night shift work** as **probably carcinogenic to humans** (Group 2A). The evidence is strongest for breast, prostate and colorectal cancer. Sleep is not wasted time: it is maintenance time.

● CAPITALISM

Jonathan Crary described contemporary capitalism as a 24/7 system that treats sleep as the last frontier that does not yet produce. Shoshana Zuboff has called **surveillance capitalism** the business model that extracts data from human behaviour to predict and modify it. Tim Wu has traced the history of the "attention merchants": from the penny press to social media, the product on sale is our eyes.

Here the metaphor becomes literal: the economy that colonises the night exposes us to a recognised cancer risk factor.

LIMITS

Not all night work can be avoided (hospitals, emergencies). The criticism targets the extension of productive time and of time captured by platforms with no social need to justify it.

SOURCES

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CENTRAL DIAGRAM

A bed at night with the moon in the window and a lit-up phone receiving notifications.

Interlude — What does it cost to be ill?

An illness is not paid for with bills alone. It is paid for with time, lost work, care, fear and interrupted plans. The medical literature already has a name for this: **financial toxicity**. In countries without universal healthcare, cancer patients who go bankrupt have a higher risk of dying. Some cancer drugs cost more than a hundred thousand dollars a year, even when the research that made them possible was largely paid for with public money. The total cost of being ill is always greater than what the accounts record, and the one who pays is not the one who profits.

Sources: Zafar, S. Y.; Abernethy, A. P. (2013), "Financial toxicity, part I", *Oncology*, 27, 80–81; Ramsey, S. D. et al. (2016), "Financial insolvency as a risk factor for early mortality among patients with cancer", *Journal of Clinical Oncology*, 34, 980–986; Experts in Chronic Myeloid Leukemia (2013), "The price of drugs for chronic myeloid leukemia (CML) is a reflection of the unsustainable prices of cancer drugs", *Blood*, 121, 4439–4442.

Part III

Cancer: Evolution, invasion and metastasis

Capitalism: Adaptation, expansion and colonisation

Common idea: the C varies, resists pressure and looks for new spaces in which to keep reproducing.

Chapter 7 — Diversity, selection and resistance

Node 37 — No C is uniform

CENTRAL MESSAGE

Within the C many different forms coexist; they compete with each other and, at the same time, cooperate against the whole.

QUESTION

Why is it so hard to fight something that changes while we watch it?

SHORT ANSWER

Because there is no single form to defeat, but a diverse population of strategies.

CORRESPONDENCE

Intratumour heterogeneity ↔ varieties of capitalism

● CANCER

In 2012 a team analysed samples from different regions of the same kidney tumour and its metastases. Each region carried different mutations: two thirds of the alterations were not shared by all regions of the tumour. A single biopsy showed only part of the story. This is

intratumour heterogeneity: a tumour is not a clone but a family tree full of branches. That is why a drug that removes one branch can let the others grow.

● CAPITALISM

Peter Hall and David Soskice distinguished **liberal market economies** (United States, United Kingdom) from **coordinated market economies** (Germany, the Nordic countries). There are also state capitalisms (China, Singapore), extractive, financial and platform capitalisms. Each combines market, state, family and finance differently.

A critique that looks at only one variety (for instance Anglo-Saxon neoliberalism) can leave the rest untouched. As in the tumour, the branches compete with each other, but they all share the trunk: private ownership of the means of production, wage labour and the obligation to accumulate.

LIMITS

Varieties of capitalism have very different outcomes in inequality, health and well-being. Recognising this does not weaken the critique: it makes it more precise.

SOURCES

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CENTRAL DIAGRAM

A tree with a single trunk and five branches: liberal, coordinated, state, financial and platform capitalism.

Node 38 — Pressure is also selection

CENTRAL MESSAGE

Every drug, every law and every crisis removes some forms of the C and leaves room for others.

QUESTION

What survives when we only attack what we already know how to see?

SHORT ANSWER

The most resistant variants, the most opaque, or those best adapted to the new rules.

CORRESPONDENCE

Therapeutic resistance through selection ↔ regulatory arbitrage and shadow banking

• CANCER

An effective treatment can wipe out 99% of a tumour. But if one cell in a million carries a mutation that makes it resistant, that cell survives and, without competition, multiplies. Months later the tumour returns, now made

almost entirely of resistant cells. Treatment did not create resistance: it **selected** it. This is one of the main reasons why chemotherapy and targeted therapies fail.

● CAPITALISM

After 2008 banks were subjected to new capital and liquidity rules (Basel III). Part of the risky activity moved to less regulated non-bank financial entities: funds, insurers, investment vehicles. Today this **non-bank financial intermediation**, also known as shadow banking, manages close to half of the world's financial assets.

The same happens with taxes: every new tax rule generates new avoidance schemes. Regulation that attacks a visible form selects the forms that know how to hide.

LIMITS

This argument does not justify giving up on regulation, any more than it justifies giving up on chemotherapy. It justifies regulating with the selection it will cause in mind.

SOURCES

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CENTRAL DIAGRAM

Three Petri dishes: one resistant cell among many, the drug leaves it alone, and months later all the cells are resistant.

Node 39 — Resistance needs no consciousness

CENTRAL MESSAGE

The C seems to learn, but only the forms that resist survive. In the case of capital, it also learns for real.

QUESTION

How can a system adapt without a single mind?

SHORT ANSWER

Through repeated selections that reward whatever keeps reproducing, and by absorbing the criticism it receives.

CORRESPONDENCE

Evolution of resistance ↔ the recuperation of critique

● CANCER

A tumour does not learn in the human sense. But under successive treatments it accumulates variants resistant to each of them. The result looks like learning: each relapse is harder to treat than the one before. Oncologists speak of **lines of treatment**: first, second, third... and each line has a lower chance of success.

● CAPITALISM

Luc Boltanski and Ève Chiapello studied French management literature from the 1960s and the 1990s. They concluded that capitalism had absorbed the **artistic critique** of May 1968 (against hierarchy, routine and alienation) and turned it into a new management model: projects, flexibility, autonomy, creativity. Critique became a slogan and work became more precarious.

Environmental movements received the same treatment: the system adopted their language and sold the solution.

LIMITS

Here the metaphor falls short: capitalism evolves not only through selection but also through conscious strategy. It has marketing departments, think tanks and consultancies devoted to learning from its critics.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a 1968 placard turned into an advertisement; on the right, bars falling from the first to the third line of treatment.

Node 40 — Camouflage in order to carry on

CENTRAL MESSAGE

The C survives when it stops looking like a threat.

QUESTION

How does it escape the defences?

SHORT ANSWER

By changing its signals, hiding information and adopting the language of normality.

CORRESPONDENCE

Immune evasion (the "don't eat me" signal) ↔ green and responsible image-washing

● CANCER

The immune system identifies abnormal cells by the molecules they display on their surface. Tumours learn to hide them: they reduce the antigen-presenting molecules (MHC class I), display the PD-L1 protein, which switches off T cells, or overexpress CD47, a signal that literally says **"don't eat me"** to macrophages. The theory of

immunoediting describes three phases: elimination, equilibrium and escape. The tumour that survives is the one that has learned to look normal.

● CAPITALISM

In 2021 a European Commission screening of environmental claims on commercial websites found that 42% were exaggerated, false or deceptive. Labels such as "sustainable", "carbon neutral" or "responsible" work like CD47: they tell the defence system there is no need to act.

The same happens with tax opacity. Gabriel Zucman estimates that about 8% of the world's household financial wealth sits in tax havens, and that 36% of multinational profits are shifted to these territories every year. Capital becomes invisible to the mechanisms that should tax it.

LIMITS

Tumour cells do not lie: they display or stop displaying molecules. Companies that greenwash do lie, which is why the European Union has passed specific rules against misleading environmental claims.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a bottle with a green leaf and an "eco" label next to a factory; on the right, a tumour cell holding up a sign that says "don't eat me".

Node 41 — What lies dormant can return

CENTRAL MESSAGE

The C can stay asleep for decades without having disappeared.

QUESTION

What remains after an apparent victory?

SHORT ANSWER

Scattered seeds waiting for favourable conditions to wake up.

CORRESPONDENCE

Dormancy of disseminated tumour cells ↔ hibernating neoliberalism (1947–1973)

● CANCER

Many tumour cells leave the original tumour very early and remain **dormant** in other organs, not dividing, invisible to tests. They can stay that way for years or decades. In hormone-receptor-positive breast cancer, the risk of relapse persists up to twenty years after diagnosis. A change in the microenvironment (inflammation, ageing, a hormonal shift) can wake them up.

● CAPITALISM

In 1947, at the height of the expansion of the welfare state, Friedrich Hayek gathered a group of economists and philosophers in Switzerland: the **Mont Pèlerin Society**. For twenty-five years their ideas (free markets, a minimal state, weak unions) were marginal. They lay dormant in universities and foundations. When the crisis of the 1970s destabilised the Keynesian consensus, those seeds woke up: Chile in 1973, the United Kingdom in 1979, the United States in 1980. Eight members of the Mont Pèlerin Society received the Nobel Prize in Economics.

LIMITS

Tumour dormancy has no strategy; neoliberalism organised itself deliberately to wait for its moment. Once again, the difference aggravates the economic case.

SOURCES

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CENTRAL DIAGRAM

Seeds asleep underground, with the dates 1947, 1973, 1979 and 1980; one of them begins to sprout.

Chapter 8 — Occupying new spaces

Node 42 — To grow is to occupy

CENTRAL MESSAGE

The C does not grow in a vacuum: every expansion displaces someone.

QUESTION

What does each new expansion displace?

SHORT ANSWER

Tissue, territory, time, autonomy or imagination.

CORRESPONDENCE

Local invasion and matrix degradation ↔ enclosures, land grabs and gentrification

● CANCER

Invasion is the step that turns a tumour into a cancer. Tumour cells manufacture enzymes, matrix metalloproteinases, that dissolve the basement membrane and the matrix that holds tissue architecture together. Then they advance through the spaces they have opened, compress the neighbouring structures and take the place of the cells that lived there.

● CAPITALISM

Marx located the origin of capitalism in the **enclosure** of common land in England: peasants driven off the land they farmed and forced to sell their labour power in the towns. He called it **primitive accumulation**. David Harvey has added that it is not a past phase but a permanent mechanism: **accumulation by dispossession**.

The examples are current: investment funds grabbing farmland in the Global South, entire neighbourhoods gentrified by tourist flats, privatised beaches, rivers and public spaces. The expansion of the **C** always has a displaced population.

LIMITS

Displaced cells cannot resist; communities can. Housing and land movements prove it every day.

SOURCES

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CENTRAL DIAGRAM

A street: small houses, a building of tourist flats and neighbours leaving with their boxes.

Node 43 — Looking for new territories

CENTRAL MESSAGE

When one space is saturated, the C looks for another.

QUESTION

What drives the flight forward?

SHORT ANSWER

The exhaustion of the place it is in: less oxygen, lower profits.

CORRESPONDENCE

Hypoxia that triggers migration ↔ the spatial fix

● CANCER

When a tumour grows faster than its vessels, its interior runs out of oxygen. This **hypoxia** activates a factor called HIF-1, which switches on programmes for survival, vessel formation and migration. Cells under hypoxia are more mobile and invasive. The exhaustion of the place pushes the tumour to look for another.

● CAPITALISM

In 1913 Rosa Luxemburg argued that capitalism needs a **non-capitalist outside** in which to sell, invest and obtain raw materials. David Harvey calls it the **spatial fix**: when too much capital accumulates in a region without enough profitable opportunities, capital moves to new territories, builds new infrastructure and opens new markets. Crises are not solved: they are moved.

The colonisation of outer space is the latest version of this flight, with the same scheme: if this body runs out, we'll find another.

LIMITS

Tumour migration is a cellular process; the relocation of capital is a corporate and political decision, and receiving populations can intervene.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a ship leaving an exhausted port; on the right, a tumour whose centre has no oxygen, with cells escaping from it.

Node 44 — Crossing borders

CENTRAL MESSAGE

The C becomes systemic when it breaks through the barriers that kept it localised.

QUESTION

What changes when it stops respecting compartments?

SHORT ANSWER

The problem stops being local: it affects the general circulation of the body or of society, and contaminates countries, classes, cultures and languages.

CORRESPONDENCE

Metastasis through blood and lymph ↔ the globalisation of capital

● CANCER

Metastasis is the process by which cells from a tumour cross the vessel wall, travel through the blood or lymph, survive the journey, exit in a distant organ and form a new tumour there. To do so, many cells switch on an embryonic programme called the **epithelial–mesenchymal transition**: they stop being anchored to their neighbours and become mobile. It is a very

inefficient process (the vast majority of circulating cells die), but it is enough that a few arrive. Metastasis is the cause of most cancer deaths.

● CAPITALISM

In 1848 Marx and Engels described it with almost clinical precision: "The need of a constantly expanding market for its products chases the bourgeoisie over the entire surface of the globe. It must nestle everywhere, settle everywhere, establish connexions everywhere."

Globalisation has not only exported goods. It has exported models of consumption, housing, food and leisure. It has homogenised cultures and sidelined languages: according to Ethnologue, about 40% of the world's more than 7,000 languages are endangered. The metastasis of capital does not respect the compartments of class, nation or culture.

LIMITS

The classic claim that 90% of cancer deaths are due to metastasis has been questioned for lack of solid data; it is better to say "most". Globalisation has also spread knowledge, medicine and rights.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a globe with routes and speech bubbles of languages fading out; on the right, a body with cells travelling through the blood.

Node 45 — Turning life into territory

CENTRAL MESSAGE

The C prepares the ground before it arrives: it turns into a commodity what was not one before.

QUESTION

What happens when caring, talking, moving or remembering become business?

SHORT ANSWER

Spheres of life that were common enter circuits of property, data and profitability.

CORRESPONDENCE

Pre-metastatic niche ↔ prior commodification (fictitious commodities)

● CANCER

In 2005 a research team discovered that tumours prepare the organs where they will metastasise **before** cancer cells arrive. They send vesicles (exosomes) and signals that modify the distant tissue, attract bone-marrow cells to it and turn it into fertile soil. This is the **pre-metastatic niche**. When tumour cells arrive, the ground is ready.

● CAPITALISM

Karl Polanyi called land, labour and money **fictitious commodities**: things not produced to be sold but treated by the market as if they were. Contemporary capitalism has added many more: personal data, attention, care for the elderly, the genome, atmospheric carbon and even water. Since December 2020 Californian water has been traded on a futures market in Chicago.

Before capital arrives, deregulation, privatisation or the creation of a new property right arrives. That is the pre-metastatic niche of the economy.

LIMITS

The pre-metastatic niche is a biochemical process; commodification requires laws, contracts and political decisions that can be challenged.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, "for sale" signs with the euro symbol over a field, and a water tap; on the right, vesicles leaving the tumour for a distant organ.

Node 46 — Organising extraction

CENTRAL MESSAGE

The C does not just occupy a territory: it reorganises it to work for it.

QUESTION

What turns expansion into domination?

SHORT ANSWER

Forcing the occupied territory to destroy itself to feed its occupier.

CORRESPONDENCE

The vicious cycle of bone metastases ↔ the colonial enclave economy

● CANCER

When breast or prostate cancer spreads to bone, the tumour cells do not destroy the bone directly. They send signals that activate osteoclasts, the bone cells responsible for resorbing it. As it dissolves, the bone releases growth factors that were stored in it, and those factors feed the tumour, which sends more signals. Oncologists call it the **vicious cycle**: the occupied tissue takes itself apart to feed its occupier.

● CAPITALISM

Colonial economies worked the same way. In Potosí, the silver mountain of the Spanish empire, thousands of indigenous people died in the mines under the forced-labour system of the *mita*. In the Congo Free State, the personal property of King Leopold II of Belgium, rubber was extracted through mutilation and hostage-taking. The occupied territories were reorganised so that their own labour and their own land fed the metropolis.

Today's **enclave economies** (mines, plantations, free-trade zones) keep the structure: the territory exports matter and imports waste.

LIMITS

This is one of the nodes where the metaphor is most exact in form and most inadequate in morality. The vicious cycle in bone has no conscious victim; colonisation had millions of dead and perpetrators with names.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, the mountain of Potosí hollowed out by a mine; on the right, a bone that dissolves and feeds the tumour in a vicious cycle.

Node 47 — Metastasis has no army

CENTRAL MESSAGE

A metastasis does not legislate, conquer or sign treaties. An empire does.

QUESTION

Which difference can the metaphor not erase?

SHORT ANSWER

The difference between a biological process without intent and organised, armed and legalised violence.

CORRESPONDENCE

Metastasis (without intent) ↔ a commercial empire (with an army, lawyers and shareholders)

● CANCER

Metastasis is devastating, but it is blind. Tumour cells do not know where they are going. Most die in the blood. The few that colonise an organ do so because the soil suits them, not because they chose it.

● CAPITALISM

The British East India Company was a private company with shareholders in London. At the start of the nineteenth century it had an army of some 200,000 soldiers, twice the size of the British army, and governed much of the Indian subcontinent. The Dutch East India Company had the right to declare war, sign treaties and mint coins. Both were listed on the stock exchange.

The conquest was planned, financed and justified with legal and religious doctrines. No metastasis has a board of shareholders.

LIMITS

This node is an explicit limit of the metaphor. It does not soften the critique: it sharpens it. What in cancer is a biological accident was, in capitalism, a business.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, an East India Company ship with a flag, soldiers and a contract; on the right, cells travelling blindly through the blood, most of them dying.

Chapter 9 — Ecology, saturation and collapse

Node 48 — Destroying what one depends on

CENTRAL MESSAGE

The C lives off the system it destroys.

QUESTION

Can it survive after exhausting its host or its planet?

SHORT ANSWER

No. Its local success contains the possibility of total failure.

CORRESPONDENCE

Death of the host and death of the tumour ↔ cannibal capitalism and the exhaustion of the conditions of production

● CANCER

Cancer is an **evolutionary dead end**. Each tumour starts from scratch in each patient and disappears with them. All the adaptations it has accumulated (resistance, invasion, metastasis) die with the body it has destroyed. From the tumour cell's point of view, it is a strategy that wins every battle and loses the only war.

● CAPITALISM

The economist James O'Connor spoke of the **second contradiction of capitalism**: the system degrades the conditions of production it did not produce itself (nature, workers' health, infrastructure). Nancy Fraser calls it **cannibal capitalism**.

The data are concrete. According to the Food and Agriculture Organization of the United Nations (FAO), a third of the world's soils are moderately or highly degraded. Industrial agriculture exhausts the soil it needs to produce, and industrial fishing empties the seas it lives from.

LIMITS

An economic system does not die like an organism; it can be transformed before it destroys its base. That is the difference that makes the rest of this book possible.

SOURCES

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CENTRAL DIAGRAM

A person sitting on the branch of a tree saws through that very branch.

Node 49 — There is no "outside"

CENTRAL MESSAGE

There is no exterior to which extraction and waste can be sent indefinitely. Everything expelled comes back.

QUESTION

Where do the expelled costs end up?

SHORT ANSWER

Inside the same system. Often, inside our own bodies.

CORRESPONDENCE

Tumour acid and toxins that stay in the body ↔ pollutants that return as carcinogens

● CANCER

A tumour's waste does not leave the body. Acid, inflammatory substances and fragments of dead cells circulate in the blood and affect distant organs: fever, fatigue, anaemia, thrombosis. These are the **paraneoplastic syndromes**: remote effects of a tumour that no one sees.

● CAPITALISM

Classical economists treated nature as an **infinite sink**: the atmosphere, rivers and oceans would absorb any waste. They do not. Since 2023 the IARC has classified perfluorooctanoic acid (PFOA), one of the "forever chemicals" of the PFAS family used in non-stick pans and waterproof fabrics, as **carcinogenic to humans** (Group 1). It is present in the blood of almost the entire population. In 2025 microplastics were measured in human brain tissue, at higher concentrations than in samples from 2016.

The waste that industry expelled into the environment returns to the body as a cancer risk factor. The metaphor closes in on itself.

LIMITS

Here there is no metaphor: the classification of PFOA establishes a causal link between an industrial product and cancer. The measurement of microplastics in human tissue, on the other hand, is still methodologically contested and does not by itself demonstrate any effect on health.

SOURCES

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CENTRAL DIAGRAM

A planet: a non-stick pan releases PFOA that returns to a person's blood.

Node 50 — The limit returns as crisis

CENTRAL MESSAGE

The C can deny limits, but it cannot abolish them.

QUESTION

How do ignored boundaries reappear?

SHORT ANSWER

As necrosis, scarcity, financial crisis, climate change and loss of function.

CORRESPONDENCE

Necrotic core ↔ recurring systemic crises

● CANCER

When a tumour grows faster than its vessels, the cells in the centre, too far from oxygen, die en masse. A **necrotic core** forms: a zone of dead tissue inside the tumour.

Necrosis does not stop the tumour, but it inflames it and often makes it more aggressive, because it selects the cells that tolerate the lack of oxygen.

● CAPITALISM

Capitalism has always lived with crises. The International Monetary Fund has counted 151 systemic banking crises worldwide between 1970 and 2017. Each crisis destroys capital, firms and jobs, and often leaves behind a more concentrated system. Ecological crises follow the same pattern: the ignored limit returns as drought, fire or crop failure.

Minsky formulated it for finance: stability breeds instability. The limit gives no warning: it explodes.

LIMITS

Some economists see crises as a cleansing function of the system. But necrosis does not cure the tumour either: it makes it more resistant.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, 151 dots, one for each systemic banking crisis between 1970 and 2017; on the right, a tumour with a dead core.

Node 51 — Local success can be global ruin

CENTRAL MESSAGE

A part can win while the whole loses its capacity to live.

QUESTION

Can a profit that destroys its own future conditions be called a victory?

SHORT ANSWER

Only if we measure the present and ignore the whole system.

CORRESPONDENCE

Short-sighted evolution of tumour cells ↔ short-termism and share buybacks

● CANCER

Biologists speak of **short-sighted evolution**: selection favours whatever gives an immediate advantage, even if it is disastrous in the long run. A tumour cell that divides faster outcompetes its neighbours inside the tumour, even though the final result is the death of the host and its own. Selection does not see the future.

● CAPITALISM

Between 2003 and 2012 the 449 companies in the S&P 500 index that were listed throughout the period spent 54% of their earnings buying back their own shares (to push up the price) and a further 37% on dividends. Only 9% was left to invest in production, research or wages. William Lazonick called it **profits without prosperity**.

The same happens on a planetary scale: to limit warming to 2 °C, about a third of oil reserves, half of gas reserves and over 80% of coal reserves would have to remain unburned. These reserves appear as assets on balance sheets. Accounting success is climate ruin.

LIMITS

Firms cannot see the future, but societies can: they can set rules that require it to be counted. The cell has no such option.

SOURCES

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CENTRAL DIAGRAM

A divided bar: 54% share buybacks, 37% dividends and 9% investment; beside it, a falling curve.

Node 52 — Diversity protects the whole

CENTRAL MESSAGE

When a single clone dominates, the whole becomes fragile.

QUESTION

What is lost when a single form dominates the whole tissue or the whole market?

SHORT ANSWER

Alternatives, redundancy, adaptability and resilience in the face of crises.

CORRESPONDENCE

Clonal haematopoiesis ↔ monopoly, monoculture and single point of failure

● CANCER

Blood is made from tens of thousands of different stem cells. With age, in many people, a single clone with an advantageous mutation takes over and comes to produce a large share of blood cells. This is **clonal haematopoiesis**. A 2014 study showed that it multiplies

the risk of leukaemia by eleven and increases the risk of cardiovascular disease. Loss of diversity is, in itself, a risk factor.

● CAPITALISM

In Ireland, dependence on a single potato variety turned a blight into the Great Famine of 1845–1849: a million dead and a million emigrants. Today most bananas exported worldwide are of the Cavendish variety, threatened by a fungus. In July 2024 a faulty update from a single cybersecurity firm crashed some 8.5 million computers and stopped airports, hospitals and banks around the world.

Andrew Haldane (Bank of England) and the biologist Robert May applied ecology to banking: a homogeneous financial system, in which all banks hold the same assets, is more fragile in the face of a shock.

LIMITS

Concentration does not always come from natural selection: it is often produced by mergers approved by regulators. Here the solution is not biological, but competition law.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a row of identical potato plants, all wilted; on the right, diverse blood cells and, below, a single clone that takes over everything.

Part IV

Cancer: Diagnosis and treatment

Capitalism: Critique and transformation

Common idea: detecting the **C** is not enough; one must intervene without destroying the tissue one wants to protect.

Chapter 10 — Prevent and diagnose

Node 53 — Acting before the damage

CENTRAL MESSAGE

Prevention begins before the C is visible, where someone manufactures the risk.

QUESTION

Where does a treatment really begin?

SHORT ANSWER

In the conditions that produce the exposure: what is sold, what is emitted, what is allowed.

CORRESPONDENCE

Primary prevention ↔ regulating the commercial determinants of health

● CANCER

Cancer prevention is one of medicine's success stories, and it was almost never medical. In 1950 Richard Doll and Austin Bradford Hill demonstrated the link between tobacco and lung cancer. Less smoking, the ban on asbestos, and the vaccines against human papillomavirus

and hepatitis B have prevented millions of cancers. The World Health Organization estimates that between 30% and 50% of today's cancers could be prevented.

● CAPITALISM

Almost every effective form of cancer prevention has collided with commercial interests. The tobacco industry denied the evidence for forty years. Asbestos went on being sold for decades after its risk was known; Spain did not ban it until 2002. According to WHO Europe, four industries (tobacco, alcohol, ultra-processed food and fossil fuels) are associated with some 2.7 million deaths a year in the European region.

Preventing cancer largely means **limiting what the market may sell**. Preventing the economic **C** means limiting concentration before it becomes political power: progressive taxes, competition law and limits on private funding of politics.

LIMITS

Not all preventable cancers depend on industries (physical inactivity, for example). But the biggest risk factors have a product and a marketing strategy behind them.

SOURCES

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CENTRAL DIAGRAM

A chain: product, exposure, damage, tumour and hospital; the prevention barrier is placed at the first step.

Node 54 — Reducing risk does not remove the cause

CENTRAL MESSAGE

Damage can be reduced without touching the mechanism that produces it. Sometimes it even gets worse.

QUESTION

What does partial prevention achieve?

SHORT ANSWER

It buys time and saves lives, but it can leave the cause intact and make it harder to see.

CORRESPONDENCE

"Light" cigarettes ↔ corporate social responsibility and carbon offsets

● CANCER

In the 1960s the tobacco industry launched filter cigarettes and "light" cigarettes, with ventilation holes in the filter. On measuring machines they gave less tar. In the lungs they did not: smokers inhaled harder and deeper. A 2017 study linked these ventilated filters to the rise of lung adenocarcinoma, a type of cancer that

appears in the peripheral areas of the lung. In 2006 a US federal court concluded that the tobacco companies had knowingly deceived consumers with "light" labels.

● CAPITALISM

Corporate social responsibility and **carbon offsets** play a similar role. A study published in *Science* in 2023 analysed forest-conservation projects that sold carbon credits and concluded that most had not significantly reduced deforestation: the credits offset real emissions with imaginary reductions.

A reform that does not touch the logic of accumulation can buy time, but it can also act as a ventilated filter: it makes consuming the product look safe.

LIMITS

Harm reduction is a legitimate public-health strategy (needle-exchange programmes, for example). The problem is when it is designed by whoever sells the risk.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a carbon-offset certificate next to a chimney that keeps smoking; on the right, a cigarette with a perforated filter and smoke reaching deeper into the lung.

Node 55 — Making the process visible

CENTRAL MESSAGE

Diagnosing means connecting signals that, taken separately, look normal.

QUESTION

What must we observe to recognise the C?

SHORT ANSWER

Concentration, inequality, inflammation, loss of function and whatever is being hidden.

CORRESPONDENCE

Pathological diagnosis and TNM staging ↔ distributional national accounts and wealth registers

● CANCER

A cancer is rarely diagnosed from a single symptom. Tiredness, weight loss, mild anaemia, a change in a mole: separately they look normal. Diagnosis connects them. Then a biopsy confirms the nature of the cells and **TNM staging** (tumour size, nodes involved, metastases) measures its extent. Without that measurement no treatment can be chosen.

● CAPITALISM

For decades economic statistics measured total growth but not who took it. Thomas Piketty, Emmanuel Saez and Gabriel Zucman have built **distributional national accounts**: they allocate the whole of GDP across the population and show what share goes to each group. These accounts show that in the United States the pre-tax income of the poorer half has barely improved since 1980.

What is not measured does not exist politically. Tax havens are places designed precisely to make diagnosis impossible. That is why a public register of the beneficial ownership of companies and assets is, literally, a diagnostic test.

LIMITS

Data transform nothing on their own. But treatment without diagnosis is a gamble in the dark.

SOURCES

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CENTRAL DIAGRAM

A corkboard with notes joined by red string: tiredness, weight loss, anaemia, inequality, inflammation, tax haven.

Node 56 — Growth does not diagnose health

CENTRAL MESSAGE

A rising figure can coexist with a body or a society that is getting worse.

QUESTION

Why do the size of a tumour or GDP explain so little on their own?

SHORT ANSWER

Because they show neither function, distribution, quality of life nor cost.

CORRESPONDENCE

Surrogate endpoints (tumour shrinkage) ↔ GDP as an indicator of well-being

● CANCER

Clinical trials often measure whether a drug **shrinks the tumour** (RECIST criteria) or delays its growth, because that is faster than waiting to learn whether patients live longer. These are **surrogate endpoints**. A 2015 review showed that their correlation with actual survival is generally low. Of the cancer drugs approved in the United

States on the basis of a surrogate endpoint, only 14% later showed that they extended life. A tumour can shrink while the patient gets worse.

● CAPITALISM

Simon Kuznets, the economist who designed national accounting, warned the US Senate in 1934 that "the welfare of a nation can scarcely be inferred from a measurement of national income". Robert F. Kennedy said it in 1968: GDP "measures everything, in short, except that which makes life worthwhile".

Between 2014 and 2017 US life expectancy fell three years in a row while GDP grew. Anne Case and Angus Deaton attributed it to **deaths of despair**: overdoses, alcohol and suicide. A war, an oil spill or an epidemic make GDP grow. GDP is capitalism's surrogate endpoint.

LIMITS

GDP was not designed to measure well-being, and its authors knew it. The problem is having turned it into the goal of economic policy.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, GDP rises while life expectancy falls; on the right, a tumour that shrinks while the patient's pulse does not improve.

Node 57 — Measure functions, not just volumes

CENTRAL MESSAGE

A healthy system is recognised by what it allows people to do and live, not by its size.

QUESTION

Which indicators should accompany growth?

SHORT ANSWER

Health, free time, distribution, autonomy, resilience, emissions and the quality of relationships.

CORRESPONDENCE

Overall survival and quality of life ↔ well-being indicators and ecological limits

● CANCER

Oncology has learned to measure what matters. **Overall survival** and **quality of life** are the reference endpoints. The European Society for Medical Oncology (ESMO) has created a scale to assess whether a drug brings real clinical benefit, not just tumour shrinkage. A treatment

that gives two extra weeks of life with heavy toxicity is not valued the same as one that gives two years in good condition.

● CAPITALISM

In 2009 a commission chaired by Joseph Stiglitz, Amartya Sen and Jean-Paul Fitoussi recommended shifting from measuring mainly production to measuring well-being: household income, distribution, health, education, care work and sustainability. Kate Raworth has drawn it as a **doughnut**: a safe space between a social foundation (what everyone needs) and an ecological ceiling (what the planet can bear). New Zealand passed its first well-being budget in 2019.

LIMITS

Changing the indicator does not in itself change ownership or power. But it decides what counts as success, and therefore what is rewarded.

SOURCES

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CENTRAL DIAGRAM

Kate Raworth's doughnut: an inner ring (social foundation), an outer ring (ecological ceiling) and, in between, the safe and just space.

Chapter 11 — Treating the focus and transforming the system

Node 58 — Cutting out the focus

CENTRAL MESSAGE

Sometimes a tumour, a monopoly or a specific practice has to be removed.

QUESTION

When is a local intervention useful?

SHORT ANSWER

When the focus is well defined and removing it prevents it from spreading. But the margins must be watched.

CORRESPONDENCE

Surgery with clear margins ↔ breaking up monopolies

● CANCER

Surgery remains the most effective treatment for many localised cancers. If they are detected early and removed with **clear margins** (a rim of healthy tissue with no tumour cells), many are cured. But if cells remain at the margins, or if some had already left before the operation, the tumour comes back. That is why surgery is often combined with additional treatments.

● CAPITALISM

In 1911 the US Supreme Court ordered Standard Oil, John D. Rockefeller's monopoly, to split into 34 companies. In 1984 the telephone monopoly AT&T was broken into seven regional companies. They were successful operations... with contaminated margins. In 1999 two of Standard Oil's heirs, Exxon and Mobil, merged again to form ExxonMobil. In 2005 one of AT&T's heirs bought the parent company and took back its name. The tumour grew back in the same place.

LIMITS

Competition law works, but only if vigilance continues after the operation. Surgery without follow-up is half surgery.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a Standard Oil barrel split in 1911 into 34 companies, two of which merge again in 1999; on the right, a scalpel cutting with a margin, and a cell that was already outside.

Node 59 — Containing is not transforming

CENTRAL MESSAGE

The C can shrink without losing the logic that reproduces it.

QUESTION

Does a reform cure or merely manage?

SHORT ANSWER

It only cures if it changes the conditions that make a return likely.

CORRESPONDENCE

Cancer stem cells that survive treatment ↔ reforms that leave ownership untouched

● CANCER

Many chemotherapies kill cells that divide rapidly. The tumour shrinks, sometimes dramatically. But an influential theory, that of **cancer stem cells**, proposes that inside the tumour there is a small population of cells that divide slowly, resist treatment and can regenerate the whole tumour. If treatment removes the mass but not the roots, relapse is only a matter of time.

● CAPITALISM

Between 1945 and 1975 Western democracies contained capitalism with highly progressive taxes (in the United States the top income-tax rate was 91% during the 1950s), strong unions and public services. Inequality fell as never before. But ownership of the means of production did not change hands. When the crisis of the 1970s came, the roots sprouted again: the top US rate is now 37%, and inequality has returned to early-twentieth-century levels.

LIMITS

The cancer stem cell hypothesis does not apply equally to all cancers and is debated. The general idea (removing the mass does not remove the capacity to regenerate it) is widely accepted.

SOURCES

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CENTRAL DIAGRAM

A plant cut at ground level with its root intact, sprouting again.

Node 60 — Changing the conditions of reproduction

CENTRAL MESSAGE

A systemic problem requires changing the soil and the flows that feed it, not just attacking its centre.

QUESTION

What does it mean to treat the whole system?

SHORT ANSWER

Changing who owns, who decides and how resources are shared.

CORRESPONDENCE

Microenvironment therapies and adaptive therapy ↔ transforming ownership and power

● CANCER

For decades oncology tried to kill every tumour cell with the maximum tolerated dose. Today there are strategies that act on the **soil**: drugs that normalise tumour vessels, immunotherapies that reactivate the body's defences, treatments that block signals from the microenvironment. The oncologist Robert Gatenby has proposed **adaptive therapy**: instead of trying to

eliminate the whole tumour (which selects resistant cells), keep it under control by adjusting the doses. In a pilot trial in metastatic prostate cancer this strategy prolonged disease control with less drug.

● CAPITALISM

A systemic transformation acts on the conditions that reproduce accumulation: ownership, decision-making power and distribution. There are partial examples. In Germany, since 1976, workers in large companies hold half the seats on supervisory boards (**co-determination**). Universal public services take health or education out of the market. Wealth and inheritance taxes limit the reproduction of great fortunes. Cooperatives change who owns.

LIMITS

None of these examples has transformed the system on its own. But they show that the conditions of capital's reproduction can be changed by law.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, an irrigation channel that reaches only one field; on the right, the channel redistributed to every field.

Node 61 — Treatment can also harm

CENTRAL MESSAGE

Every systemic intervention can injure healthy parts. And some "treatments" have been worse than the disease.

QUESTION

How do we transform without producing a new C?

SHORT ANSWER

With evidence, proportionality, protection of the vulnerable, the possibility of correction and democratic control.

CORRESPONDENCE

Overtreatment (Halsted radical mastectomy) ↔ shock therapy and forced collectivisation

● CANCER

For almost a century breast cancer was treated with the **Halsted radical mastectomy**: removing the breast, the chest muscles and the armpit lymph nodes. The more radical, the more effective, it was believed. Bernard Fisher's clinical trials showed otherwise: far less

aggressive surgery gave the same survival. Thousands of women had been mutilated for no benefit. The history of oncology is full of treatments that did more harm than good until a trial proved it.

● CAPITALISM

In the 1990s Russia applied **shock therapy** to move to capitalism: rapid mass privatisation, price liberalisation and cuts. Male life expectancy fell by about six years between 1990 and 1994. A study published in *The Lancet* linked mass privatisation to the rise in mortality. The treatment that killed was capitalism itself.

History also records the opposite case: the forced collectivisation of Soviet agriculture caused the famine of 1932–1933 and millions of deaths. Transformations that concentrate all power in a single centre can reproduce the **C** under another name.

LIMITS

This node does not equate all transformations: it recalls that evidence-based medicine was born precisely to avoid heroic, useless treatments. Politics needs the same principle.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, the fall in Russian life expectancy between 1990 and 1994; on the right, radiotherapy beams aimed at the tumour that avoid the healthy organs.

Node 62 — Who pays for the transition?

CENTRAL MESSAGE

The costs of a treatment are always shared out. The question is who pays them.

QUESTION

Who decides what is reduced, what is protected and who waits?

SHORT ANSWER

A public decision that prevents the same people from always paying.

CORRESPONDENCE

Financial toxicity of treatment ↔ just transition

● CANCER

Cancer treatment has side effects that are physical and also economic: lost working days, travel, carers who give up their jobs, drugs that are not covered. The medical literature calls this **financial toxicity**, and it hits hardest those who already had fewer resources. Universal health systems reduce it by spreading the cost across the whole population.

● CAPITALISM

In 2018 the French government raised the fuel tax to reduce emissions. Those who paid most were working-class people in rural areas who depended on their cars, while the government abolished the wealth tax. The response was the **yellow vests**. The lesson is clear: an ecological transition that loads the costs onto the usual people fails politically and is unjust.

In 2015 the International Labour Organization adopted guidelines for a **just transition**: protection for workers in sectors that must shrink, training, income guarantees and a say in decisions.

LIMITS

Sharing costs fairly does not make them disappear. But it decides whether a transformation has support or becomes a privilege paid for by others.

SOURCES

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CENTRAL DIAGRAM

A fuel pump with an added tax and a yellow vest.

Chapter 12 — Defences, power and relapse

Node 63 — Defence is a collective capacity

CENTRAL MESSAGE

A body protects itself from the C with distributed defences, not a single guardian.

QUESTION

Which structures can defend common life?

SHORT ANSWER

Independent science, unions, public services, organised communities, journalism and democratic institutions.

CORRESPONDENCE

Immunosurveillance ↔ countervailing power

● CANCER

In the 1950s Macfarlane Burnet and Lewis Thomas proposed the theory of **immunosurveillance**: the immune system constantly detects and eliminates transformed cells before they form a tumour. For decades it was debated; today it is well established. People with a weakened immune system have a higher risk of several cancers. In 2018 James Allison and Tasuku Honjo

received the Nobel Prize for the **immunotherapies** that release the brakes of the immune system so that it attacks the tumour again.

● CAPITALISM

In 1952 John Kenneth Galbraith described **countervailing power**: the power of big firms is only limited when other organised powers stand opposite them (unions, cooperatives, the state). When these defences weaken, inequality grows. A 2021 study of the United States in the twentieth century shows that the fall in union membership explains a significant part of the rise in income inequality. Across the OECD, union membership has roughly halved since the mid-1970s.

A **C** growing in a society without defences is like a tumour in an immunosuppressed body.

LIMITS

Social countervailing powers have interests of their own and can be corporatist or exclusionary. The immune system can also err. No defence is perfect; all are necessary.

SOURCES

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CENTRAL DIAGRAM

A megaphone, a newspaper, a flask, a hospital and a lymphocyte surround the C.

Node 64 — Defence also needs limits

CENTRAL MESSAGE

A defence without brakes can attack the very body it wants to protect.

QUESTION

How do we keep the cure from turning into authoritarianism?

SHORT ANSWER

By limiting the power of those who defend, protecting dissent and reviewing every decision.

CORRESPONDENCE

Immune-related adverse events and autoimmunity ↔ authoritarianism in the name of the common good

● CANCER

Immunotherapies that release the brakes of the immune system are very effective in some cancers, but they come at a price. The brakes were there for a reason: to stop the immune system attacking the body itself. Some treated patients suffer **immune-related adverse events**: inflammation of the colon, liver, lungs, thyroid or heart,

sometimes severe. CAR-T cell therapies can cause a potentially fatal **cytokine storm**. A runaway defence becomes autoimmune disease.

● CAPITALISM

Twentieth-century history offers examples of movements born to defend workers against capital that ended up building police states persecuting those same workers. Rosa Luxemburg, a revolutionary and critic of capitalism, warned the Bolsheviks in 1918: "Freedom only for the supporters of the government... is no freedom at all. Freedom is always and exclusively freedom for the one who thinks differently."

A critique of the **C** that does not limit its own power risks becoming what it fought.

LIMITS

This node does not equate democratic defences with authoritarianism. It states that every defence needs checks: judicial independence, press freedom and minority rights.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a speech bubble with Rosa Luxemburg's words on freedom; on the right, lymphocytes attacking an organ of the same body.

Node 65 — Changing the managers does not change the metabolism

CENTRAL MESSAGE

The C survives governments, leaders and reforms that do not touch its pipes.

QUESTION

How do we tell a replacement from a transformation?

SHORT ANSWER

By checking whether the circuits of ownership, accumulation and power have changed, not just the names.

CORRESPONDENCE

Clonal replacement after treatment ↔ alternating governments with the same ownership structure

● CANCER

When a treatment removes the dominant clone of a tumour, another clone often takes its place. The new clone may carry different mutations, but it shares the same hallmarks: it proliferates, invades and evades the

defences. This is **clonal replacement**. The patient sees the tumour markers change; the tumour goes on doing the same thing.

● CAPITALISM

"If we want things to stay as they are, things will have to change," says a character in *Il Gattopardo* (1958) by Giuseppe Tomasi di Lampedusa. Alternation between centre-right and centre-left governments in many democracies has changed the rhetoric and some social policies, but not the general direction since the 1980s: privatisation, financial deregulation, tax cuts for capital. The political scientist Colin Crouch called it **post-democracy**: elections continue, but the important decisions are taken elsewhere.

LIMITS

Not all changes of government are equivalent: they affect rights, wages and services. The critique points to the difference between changing the management and changing the structure.

SOURCES

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CENTRAL DIAGRAM

Two portraits of different rulers above; below, the same pipes of ownership.

Node 66 — Remission is not cure

CENTRAL MESSAGE

The disappearance of symptoms does not mean the mechanism has disappeared.

QUESTION

What remains after a visible improvement?

SHORT ANSWER

Risk, memory, latent structures and the need for vigilance.

CORRESPONDENCE

Minimal residual disease ↔ regulations dismantled once the danger was no longer visible

● CANCER

Imaging tests cannot see tumours smaller than about a centimetre. A cubic centimetre of tumour contains between a hundred million and a billion cells, depending on the tissue. A patient can be in **complete remission** (no evidence of disease) and still carry millions of tumour cells. Haematologists have developed **minimal residual**

disease tests capable of detecting one leukaemic cell among a hundred thousand or a million. That is why remission is monitored for years.

● CAPITALISM

After the 1929 crash the Glass-Steagall Act separated commercial from investment banking. For more than fifty years there was no major banking crisis in the United States. Precisely for that reason it was considered unnecessary and repealed in 1999. The relapse came in 2008. The same happened with capital controls, progressive taxes and rent regulation: because they worked, they came to seem superfluous.

The remission of the economic **C** in the post-war years was not a cure: it was a period below the detection threshold.

LIMITS

No system can be watched forever with the same intensity. But dismantling a protection because it works is confusing remission with cure.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, the timeline of the Glass-Steagall Act (1933–1999) and the 2008 crisis; on the right, a clean CT scan and a microscope that still finds cells.

Interlude — Side effects

No transformation is innocent. We must always ask who gains time, who loses security, who bears the effort and who can correct the decision. Medicine learned this through clinical trials, informed consent and ethics committees, after doing a great deal of harm with good intentions. Politics needs the same tools. Curing without preserving rights, pluralism and dignity can replace one **C** with another.

Part V

Cancer: Regeneration and tissue health

Capitalism: Transition and common life

Common idea: health is not stillness; it is deciding what must grow, what must shrink and what must end.

Chapter 13 — Making life grow

Node 67 — Growing with a purpose

CENTRAL MESSAGE

Not all growth is the C. The body knows how to grow as far as needed and then stop.

QUESTION

What deserves to be fed?

SHORT ANSWER

Whatever expands health, care, knowledge, autonomy and the capacity to regenerate.

CORRESPONDENCE

Liver regeneration (the hepatostat) ↔ selective growth geared to needs

● CANCER

The liver is one of the most beautiful examples of healthy growth. If two thirds of it are removed, what remains regenerates until it recovers its original mass within a few weeks. And then it stops. Not one gram more.

Biologists speak of a **hepatostat**: a system that adjusts the size of the liver to the needs of the body. It is the same

organ, with the same cells able to divide, but with one essential difference from cancer: its growth has a purpose and an end.

● CAPITALISM

Advocates of **degrowth** do not argue that nothing should grow. They argue for reducing the material and energy throughput of rich countries while growing renewable energy, public transport, healthcare, education, care and ecological restoration. Jason Hickel sums it up: the question is not "growth or no growth", but **growth of what, and for whom**.

An economy that worked like the liver would know how to grow where something is missing and stop when there is enough.

LIMITS

Deciding what should grow is a political conflict, not a biological rule. But the liver shows that growth with a limit is not a utopia: it is how every healthy organism normally works.

SOURCES

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CENTRAL DIAGRAM

A liver from which two thirds have been removed regenerates to its original size, and stops.

Node 68 — Learning to have enough

CENTRAL MESSAGE

The opposite of the C is not poverty, but shared sufficiency.

QUESTION

When can a system say it has enough?

SHORT ANSWER

When all its parts can live with dignity without sacrificing others.

CORRESPONDENCE

The Hippo signalling pathway (organ size control) ↔ sufficiency and decent living standards

● CANCER

How does an organ know it has reached its size? One of the answers is the **Hippo pathway**, a signalling system that cells use to sense the density and tension of the tissue. When the organ is big enough, the Hippo pathway switches on and halts proliferation. When this pathway fails, organs overgrow and tumours appear. The body has, literally, a molecular mechanism for saying **enough**.

● CAPITALISM

In 2020 a study calculated the energy needed for the entire world population, some 10 billion people in 2050, to live decently: adequate housing, food, healthcare, education, mobility and communications. The answer was roughly 60% less energy than the world consumes today. The problem is not that there is not enough: it is that the system has no mechanism for saying **enough**.

Aristotle knew it: the wealth needed to live well has a limit. Chrematistics does not.

LIMITS

"Enough" is not a fixed figure: it changes with culture, technology and history. But a system that cannot ask the question will never be able to answer it.

SOURCES

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- Millward-Hopkins, J. et al. (2020). "Providing decent living with minimum energy: a global scenario". *Global Environmental Change*, 65, 102168.
- Aristotle. *Politics*, book I.

CENTRAL DIAGRAM

A jug filling up to a line marked "enough".

Node 69 — Growing care and knowledge

CENTRAL MESSAGE

What sustains life most is what the C considers a cost.

QUESTION

What needs more resources to sustain life?

SHORT ANSWER

Health, education, care, science, culture, decent housing and ecological restoration.

CORRESPONDENCE

Tissue maintenance (stem cells and niches) ↔ Baumol's "cost disease"

● CANCER

An adult body devotes a great deal of energy to maintaining itself, not to growing. Stem cells in the gut, skin and blood constantly replace worn-out cells; the immune system keeps watch; DNA-repair mechanisms correct thousands of errors in every cell every day. This maintenance work does not make the body grow, but without it the body would collapse within weeks.

● CAPITALISM

In 1967 the economist William Baumol explained why healthcare, education or the arts seem to get ever more expensive: they cannot raise productivity like a factory. A Beethoven string quartet needs the same musicians as in 1800, and a nurse cannot look after twice as many patients without worsening the care. He called it **cost disease**.

The name itself is a diagnosis of the **C**: the system calls "disease" the work that keeps society alive, because it cannot be sped up. That is why it is the first to be cut and the worst paid.

LIMITS

Baumol did not say these sectors were dispensable; on the contrary, he argued that a richer society can afford them. The problem is the logic that treats them as expenditure rather than foundation.

SOURCES

- Weinberg, R. A. (2014). *The Biology of Cancer*, chapter 12 (maintenance of genomic integrity).
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- Baumol, W. J. (2012). *The Cost Disease: Why Computers Get Cheaper and Health Care Doesn't*. New Haven: Yale University Press.

CENTRAL DIAGRAM

Diptych. A string quartet in 1800 and the same quartet in 2026: it still takes the same four musicians.

Node 70 — Shrinking what destroys

CENTRAL MESSAGE

The transition also requires reducing harmful activities. Sometimes the best way is not to destroy them but to convert them.

QUESTION

What must stop growing, and how?

SHORT ANSWER

Fossil extraction, obsolescence, speculation, armaments and unnecessary material consumption; through conversion, not abandonment.

CORRESPONDENCE

Differentiation therapy (acute promyelocytic leukaemia) ↔ industrial conversion and just transition

● CANCER

Acute promyelocytic leukaemia was one of the deadliest leukaemias. In the 1980s haematologists in Shanghai tried a different treatment: all-trans retinoic acid, a derivative of vitamin A. It did not kill the leukaemic cells: it **forced them to mature**. Cells stuck in

an immature, proliferating state turned into functional white blood cells and died naturally. Combined with arsenic trioxide, this treatment now cures more than 90% of patients. It is **differentiation therapy**: turning what proliferates without function into something useful.

● CAPITALISM

In 2024 world military spending reached 2.7 trillion dollars, the highest ever recorded. In 2021 the International Energy Agency concluded that reaching net-zero emissions by 2050 required no new oil or gas fields. The sectors that must shrink have workers, know-how and infrastructure.

Differentiation therapy offers a model: do not abandon these sectors, convert them. In 2018 the Spanish government and the unions signed an agreement to close the coal mines with early retirement, training and development plans for the mining districts.

LIMITS

No conversion is as clean as a cell that matures: there are losses, conflicts and territories left behind. But biology shows that eliminating is not the only way to shrink.

SOURCES

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CENTRAL DIAGRAM

Diptych. On the left, a coal mine converted into a wind turbine; on the right, immature leukaemic cells that mature into white blood cells.

Chapter 14 — Sharing power and guaranteeing life

Node 71 — Decommodifying the essentials

CENTRAL MESSAGE

The body does not sell oxygen to any organ. What we need to live should not depend on the ability to pay.

QUESTION

What must remain outside the market?

SHORT ANSWER

Health, education, water, basic energy, housing and the other material conditions of life.

CORRESPONDENCE

Blood distributed according to need ↔ decommodification of essential goods

● CANCER

No organ pays to receive blood. The body distributes it according to need: during exercise, more flow to the muscles; during digestion, more to the gut; in shock, the brain and heart get absolute priority. This **autoregulation** of blood flow is the opposite of a market.

The tumour, by contrast, captures resources according to its ability to build vessels, not according to any need of the whole.

● CAPITALISM

The sociologist Gøsta Esping-Andersen called **decommodification** the capacity of a welfare state to let people live decently without depending on the market. In health, the difference is measured in lives: in the United States, cancer patients without health insurance are diagnosed later and have lower survival than insured patients. In Spain, with universal public healthcare, cancer treatment is not billed to the patient, although the illness still has economic costs: lost work, care, travel.

LIMITS

Public services can also be inefficient, bureaucratic or insufficient. But the criterion of access (need or ability to pay) is a political decision, not a technical one.

SOURCES

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- Ward, E. et al. (2008). "Association of insurance with cancer care utilization and outcomes". *CA: A Cancer Journal for Clinicians*, 58, 9–31.

CENTRAL DIAGRAM

A heart sending blood to the brain, lungs, muscles and gut; the euro sign,

crossed out.

Node 72 — Redistributing ownership and decision

CENTRAL MESSAGE

Redistributing income is not enough if power remains concentrated.

QUESTION

What must be shared besides money?

SHORT ANSWER

Ownership, information, time, decision-making power and control of infrastructure.

CORRESPONDENCE

Distributed control in the body (no cell is in charge) ↔ economic democracy

● CANCER

In a healthy organism no cell rules the others. Even the brain depends on the signals it receives from the whole body, and the immune system works without any command centre: millions of cells take local decisions coordinated by shared signals. Biological power is **distributed**. A tumour is precisely a part that has stopped listening to the common signals and imposes its own.

● CAPITALISM

Welfare states have redistributed income through taxes and benefits, but they have not redistributed ownership or decision-making power within firms. The political scientist Isabelle Ferreras proposes **economic bicameralism**: firms with two chambers, one for shareholders and one for workers, with major decisions requiring the agreement of both. Thomas Piketty proposes capping the voting power of large shareholders and creating a universal inheritance funded by progressive wealth taxes.

LIMITS

Economic democracy has coordination costs and does not by itself guarantee good decisions. But concentration does not guarantee them either, and it also excludes the majority.

SOURCES

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- Ferreras, I. (2017). *Firms as Political Entities: Saving Democracy through Economic Bicameralism*. Cambridge: Cambridge University Press.
- Piketty, T. (2019). *Capital et idéologie*, chapter 17.

CENTRAL DIAGRAM

Two semicircular chambers: the shareholders' and the workers', which must reach an agreement.

Node 73 — Combining commons, cooperatives and public services

CENTRAL MESSAGE

There is no single way to organise a healthy tissue. The body uses several at once.

QUESTION

What can each model contribute?

SHORT ANSWER

The commons, shared management; cooperatives, democratic control; public services, universality; all together, limits on private appropriation.

CORRESPONDENCE

Complementary regulatory systems (nervous, endocrine, immune) ↔ a plural economy

● CANCER

The body is not regulated by a single system. The **nervous system** is fast and precise; the **endocrine** system, slow and global; the **immune** system, distributed

and adaptive. Each compensates for the limitations of the others, and they talk to each other. A body with only one regulatory system would be far more vulnerable.

● CAPITALISM

Elinor Ostrom received the Nobel Prize in Economics in 2009 for showing that many communities manage common goods (forests, fisheries, irrigation systems) sustainably for centuries, without privatising or nationalising them. Worker cooperatives, such as those of the Mondragon group in the Basque Country, which employ some 70,000 people, show that firms can be owned by those who work in them. Public services guarantee universal access. Erik Olin Wright proposed combining these forms to erode the dominance of capital step by step.

LIMITS

Cooperatives can also reproduce the competitive logic of the market, and commons can be exclusionary. No model is a vaccine; diversity is the protection.

SOURCES

- Guyton, A. C.; Hall, J. E. *Textbook of Medical Physiology*, chapters on nervous and endocrine regulation.
- Ostrom, E. (1990). *Governing the Commons*.
- Wright, E. O. (2010). *Envisioning Real Utopias*.

CENTRAL DIAGRAM

Three structures: a communal well, a cooperative and a public hospital.

Chapter 15 — Health, conflict and maintenance

Node 74 — Cooperating is not making uniform

CENTRAL MESSAGE

A healthy tissue does not eliminate conflict or difference: it prevents one part from dominating the others.

QUESTION

How can a whole exist without a dominant part?

SHORT ANSWER

By distributing power, negotiating disagreements and preventing permanent domination.

CORRESPONDENCE

Cell competition as quality control ↔ agonistic pluralism

● CANCER

Healthy tissues are not peaceful. Cells compete constantly: **cell competition** removes damaged or less fit cells and maintains tissue quality. The immune system attacks, but it also tolerates. Health is not the absence of conflict but a regulated conflict that prevents any clone from taking over. Cancer is what happens when that balance breaks in favour of a single part.

● CAPITALISM

The philosopher Chantal Mouffe distinguishes **antagonism** (the other is an enemy to be destroyed) from **agonism** (the other is a legitimate adversary with whom one competes within shared rules). A healthy democracy does not eliminate conflicts of interest between capital and labour, between generations or between territories: it institutionalises them. The problem with capitalism is not that there is conflict, but that one side has accumulated enough power to write the rules of the conflict.

LIMITS

Biology has no legitimate adversaries and no rights. The comparison helps us see that cooperation does not require uniformity, but the rules of social conflict must be decided democratically.

SOURCES

- Vishwakarma, M.; Piddini, E. (2020). "Outcompeting cancer". *Nature Reviews Cancer*, 20, 187–198.
- Mouffe, C. (2000). *The Democratic Paradox*. London: Verso.

CENTRAL DIAGRAM

Neurons, muscle cells, red blood cells and lymphocytes, all different, connected without any one of them occupying the centre.

Node 75 — There is no single, final cure

CENTRAL MESSAGE

Health, in a body or a society, is a practice of testing, correcting and watching.

QUESTION

When can we stop watching the system?

SHORT ANSWER

Never entirely, because any solution can fail, decay or concentrate power.

CORRESPONDENCE

Cancer as a controlled chronic disease ↔ gradual, experimental transformation

● CANCER

Some cancers that used to be fatal are now chronic diseases. Before 2000, chronic myeloid leukaemia had a median survival of about five years. With imatinib and the drugs that followed, the life expectancy of many patients is approaching that of the general population. They are not cured in the strict sense: they take a daily pill and have regular check-ups. They live with the C

under control. Modern oncology has learned that it is not always necessary to eliminate everything; sometimes it is enough to keep it from dominating again.

● CAPITALISM

Karl Popper advocated **piecemeal social engineering**: reforms that can be tested, evaluated and corrected, rather than grand utopian plans that cannot be undone. John Dewey saw democracy as a continuous experiment. From a different tradition, Erik Olin Wright spoke of **interstitial and symbiotic** transformations that grow alternatives within the system until the system changes.

LIMITS

The gradual approach can also serve as an excuse to change nothing. Permanent vigilance makes sense if it is aimed at a goal, not if it replaces the goal.

SOURCES

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- Popper, K. R. (1945). *The Open Society and Its Enemies*. London: Routledge.
- Dewey, J. (1927). *The Public and Its Problems*. New York: Henry Holt.
- Wright, E. O. (2010). *Envisioning Real Utopias*.

CENTRAL DIAGRAM

A weekly pill organiser surrounded by a cycle: observe, act, evaluate and correct.

Node 76 — The common body decides

CENTRAL MESSAGE

Health begins when the whole governs its flows, its limits and its priorities again.

QUESTION

Can growth last when it has no purpose other than growing?

SHORT ANSWER

No. So the final question is not how to grow more, but who decides what we grow for and when we have enough.

CORRESPONDENCE

Homeostasis (the internal environment regulated by the whole) ↔ democratic re-embedding of the economy

● CANCER

In 1865 Claude Bernard wrote that the stability of the **internal environment** is the condition of free life. Walter Cannon called it **homeostasis**: the body's capacity to keep temperature, sugar, oxygen and thousands of variables within limits compatible with life, thanks to the

cooperation of all the organs. Cancer is, at bottom, a breakdown of homeostasis: a part that leaves common regulation. Cure, when it comes, is the return of the part to the rules of the whole.

● CAPITALISM

Karl Polanyi ended *The Great Transformation* with a proposal: to **re-embed** the economy in society, so that the market serves human needs and not the other way round. Elinor Ostrom added that this governance can be **polycentric**: many scales and many institutions that check one another.

This book began with a question: can growth last when it has no purpose other than growing? Cancer answers no. Capitalism has not answered yet. No mutation and no market will give the answer: the common body will, if it decides to.

LIMITS

The metaphor ends here. A body cannot decide about its homeostasis; a society can. That is the difference that makes the comparison not a sentence, but a warning.

SOURCES

- Bernard, C. (1865). *Introduction à l'étude de la médecine expérimentale*. Paris: J.-B. Baillière.
- Cannon, W. B. (1932). *The Wisdom of the Body*.
- Polanyi, K. (1944). *The Great Transformation*, chapter 21.
- Ostrom, E. (2010). "Beyond markets and states: polycentric governance of complex economic systems". *American Economic Review*, 100, 641–672.

CENTRAL DIAGRAM

Diptych. On the left, an assembly in a circle taking a decision; on the right, a body regulating temperature, oxygen, sugar and pH.

Summary of the architecture

1. Cellular deregulation / Deregulated accumulation

How a normal function becomes an obligation to grow without purpose.

2. Tumour microenvironment / Infrastructure of accumulation

The C needs flows, energy, labour, care, protection and consent.

3. Evolution, invasion and metastasis / Adaptation, expansion and colonisation

Diversity, resistance, occupation of new spaces and ecological crisis.

4. Diagnosis and treatment / Critique and transformation

Prevention, local intervention, systemic change, defences and relapse.

5. Regeneration and tissue health / Transition and common life

What must grow, what must shrink and how power is shared.